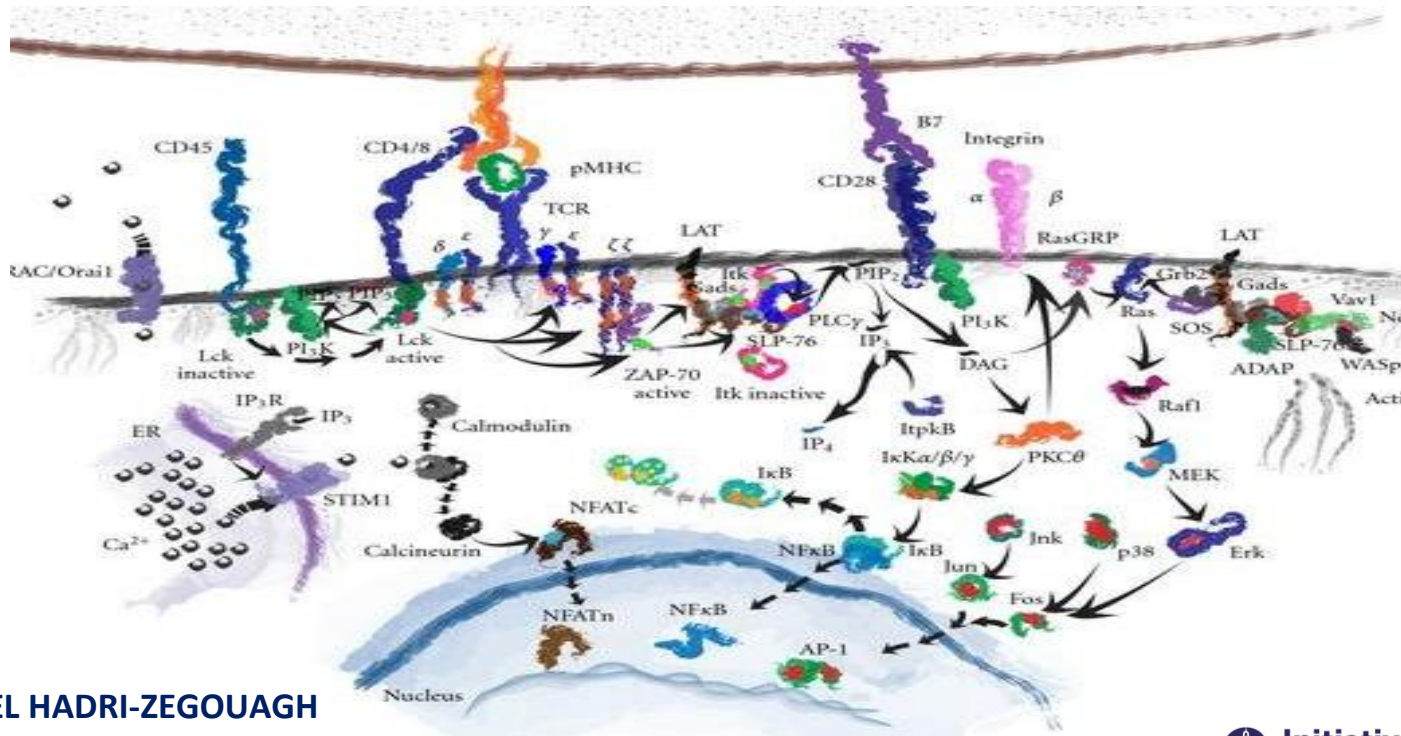


Molecular Mechanisms of Cell Signaling

(MMSC)

07 / 09 / 2026

General Principles of cell signaling



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UMR8263, CNRS-SU-U1345 Inserm

Dev2A "Développement, Adaptation et Vieillesse"

Equipe : Vieillesse cellulaire Intégratif et Inflammation

Nutrition
Qualité - Santé

- Initiale
- Apprentissage

**Initiative
Alimentation**
ALLIANCE SORBONNE UNIVERSITÉ

**ASU-GHI
Global Health Institute**
ALLIANCE SORBONNE UNIVERSITÉ

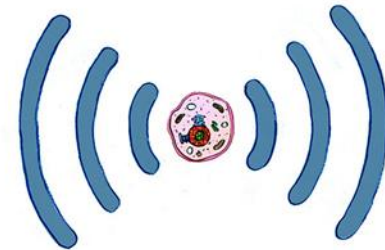
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II/ Actors of cell signal transduction

1/ Forms of chemical signaling

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2.2- Signaling proteins

2.2.1/ *adaptor proteins*

2.2.2/ *Proteins with enzymatic activity*

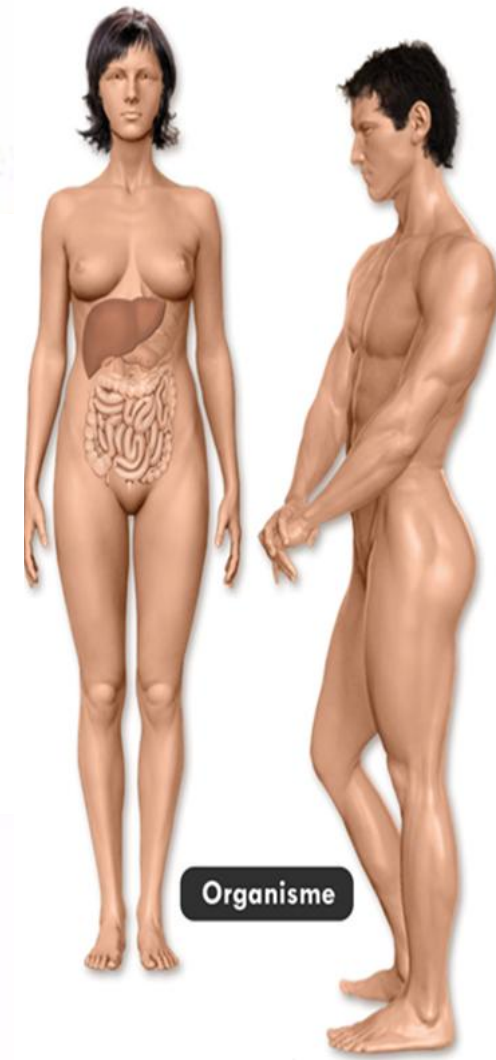
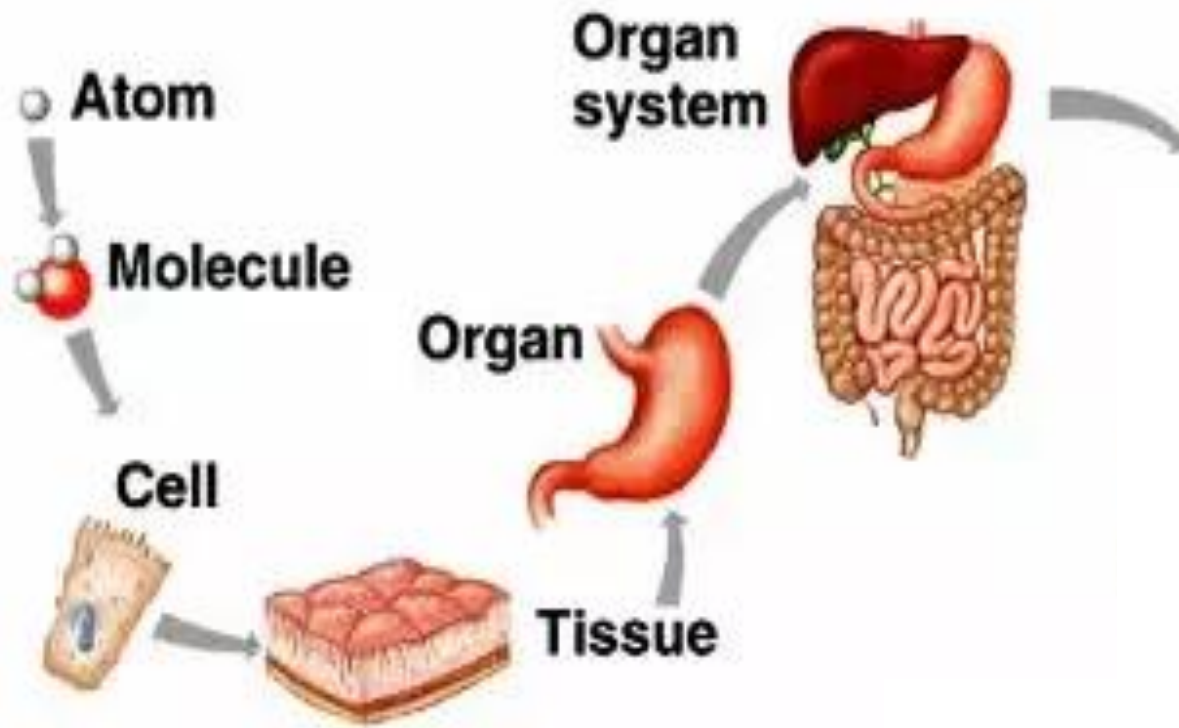
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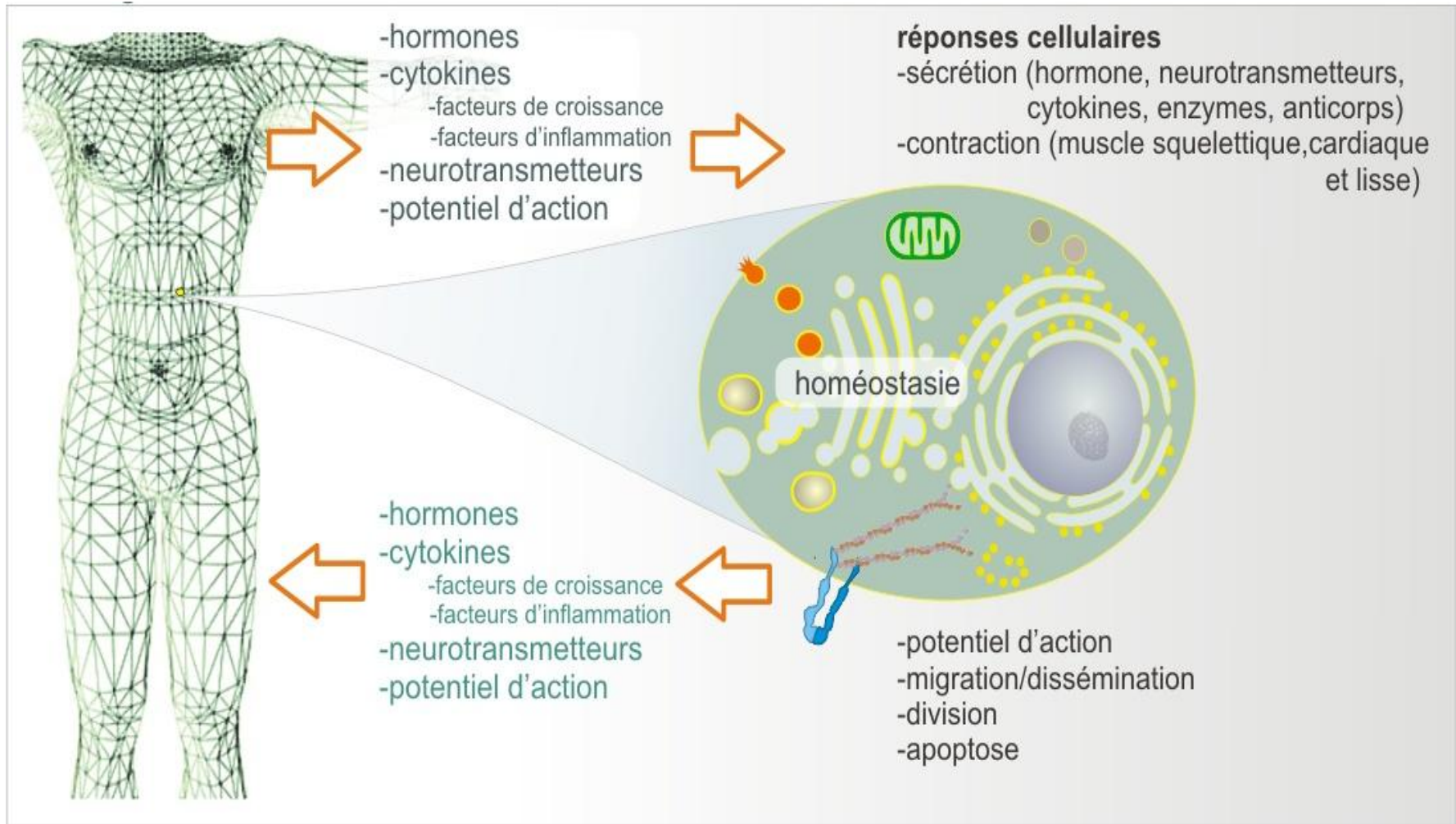
3/ Cell signaling, a Complex but not complicated network

III/ cell signaling and pathologies

From molecules to organism



Why Cells Communicate ?



**Your body is the
most complex
communication
network on earth.**

**Communication is
vital:
Within the cells,
between cells and
body systems.**



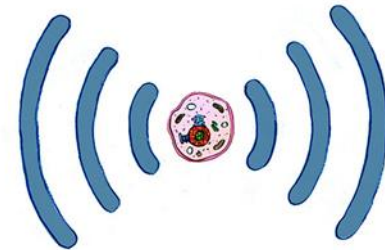
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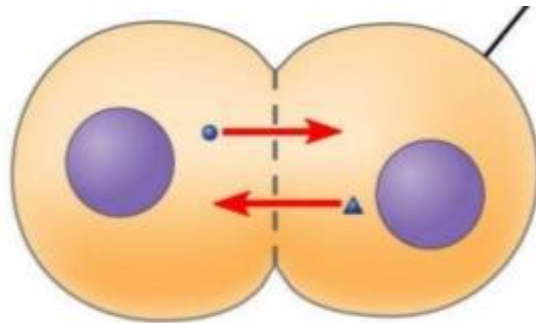
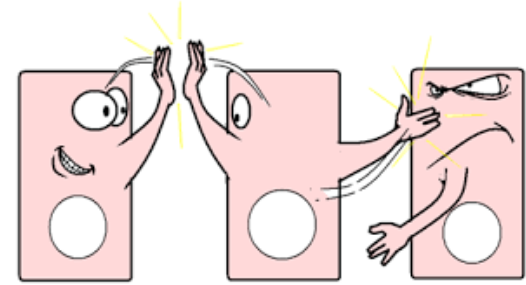
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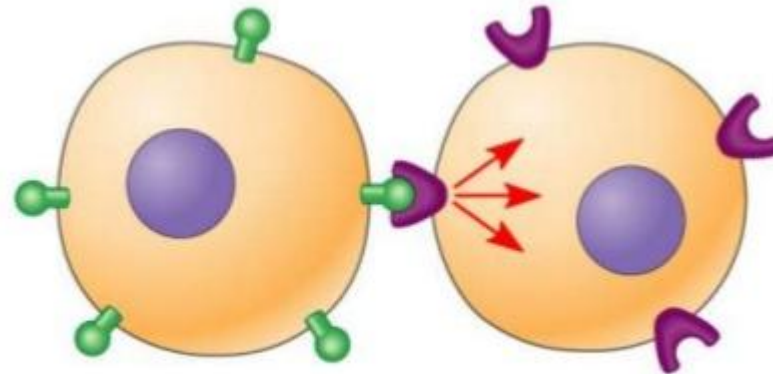
Forms of chemical signaling

Direct communication



Gap junctions
between animal cells

(a) Cell junctions



(b) Cell-cell recognition

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Direct communication allows a group of cells to coordinate their response to a signal that only one of them may have received.

Indirect communication

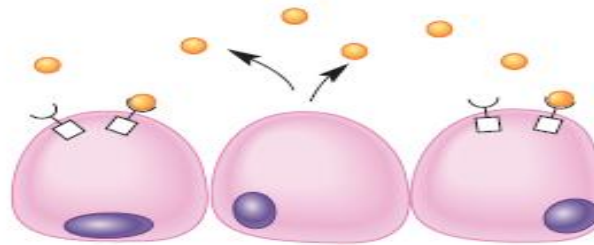
Indirect communication always involves the release of the signal

Indirect

(Travel in interstitial fluid)

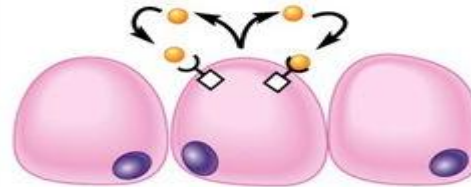
Paracrine

A cell targets a nearby cell.



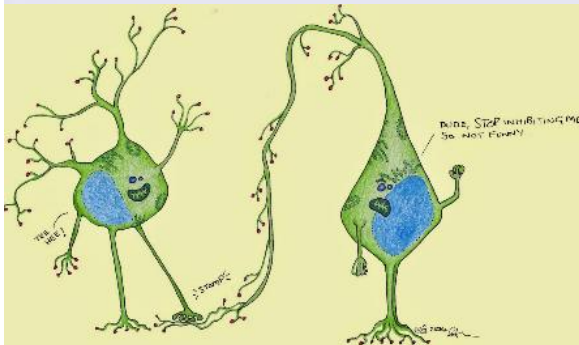
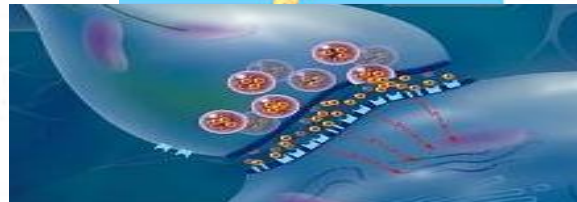
Autocrine

A cell targets itself.



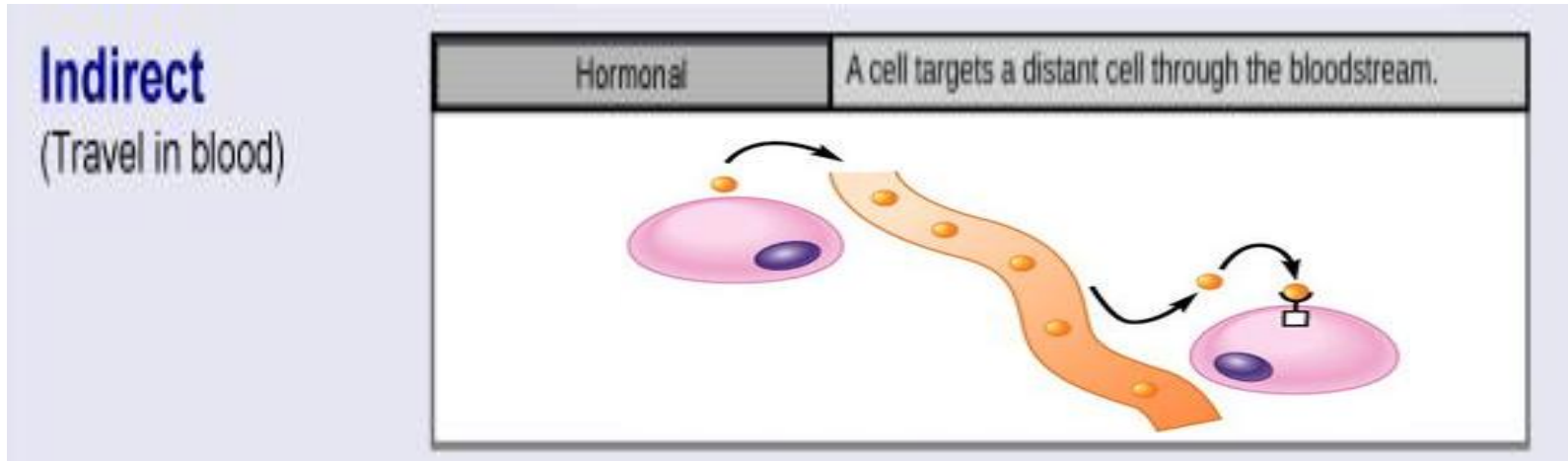
Neurotransmitter

Released in synapse close to target cell



Indirect communication

Indirect communication always involves the release of the signal

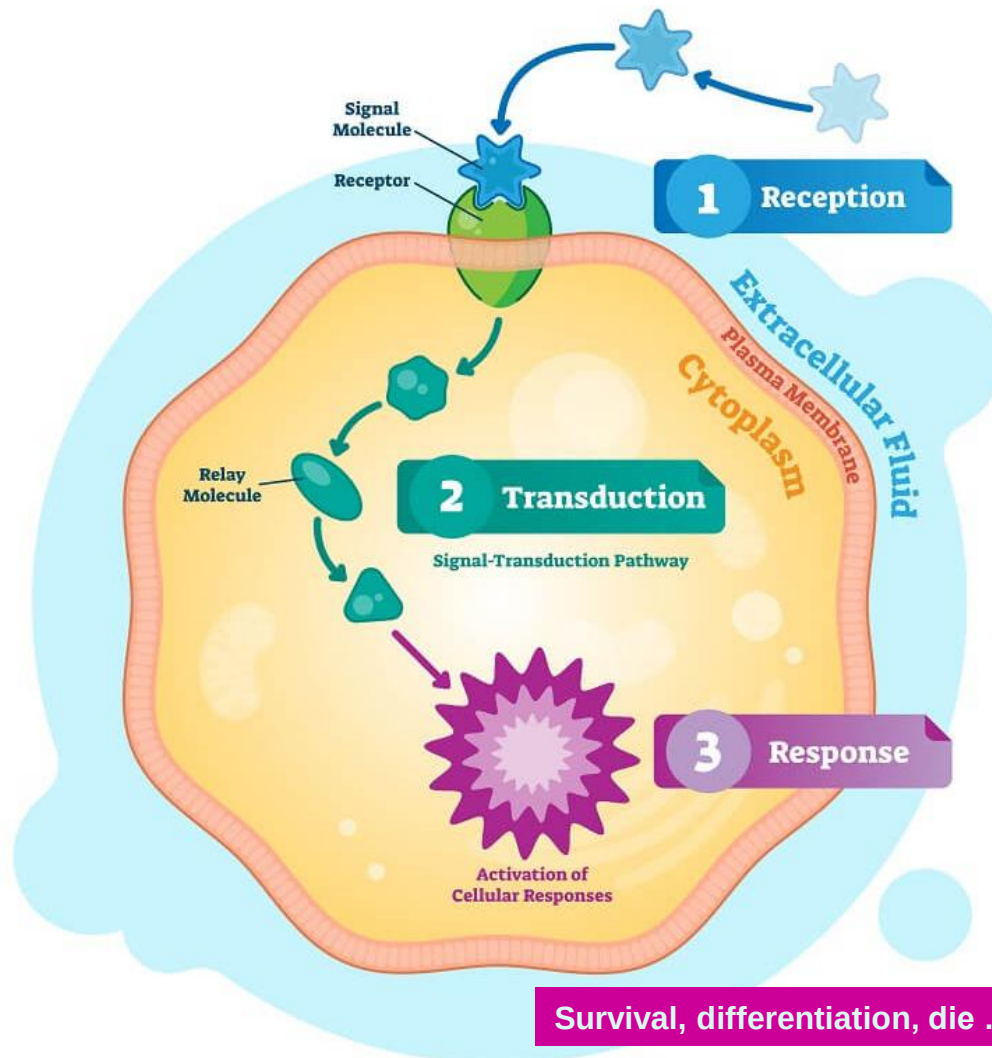


- **Hormones** are ligands that reach their targets by traveling in the blood.
- In the body, many hormones originate from **endocrine cells**.
- Endocrine cells are located in **endocrine glands** (thyroid gland, pancreas...).
- Cells in **non-endocrine organs**, such as the heart, muscle, and adipose tissue, also release hormones into the blood.

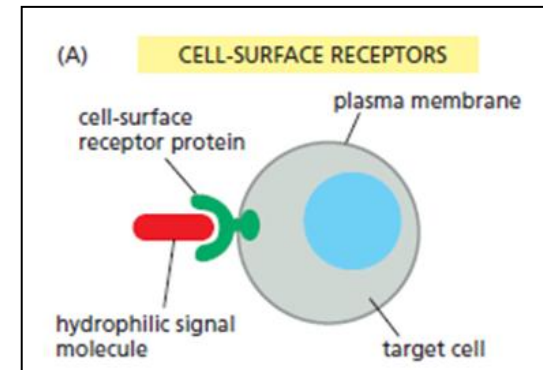
Hormonal communication can target cells that are a great distance from the signaling cells and a given signal can affect many different types of cells simultaneously.

Indirect communication

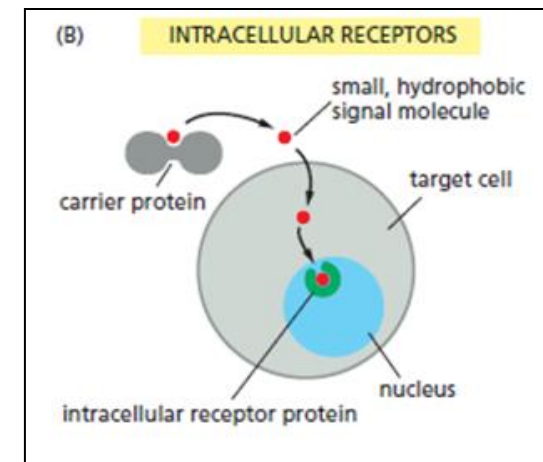
Communication between two cells involves three stages:



Receptors : Reception of the signal from the endocrine cell:



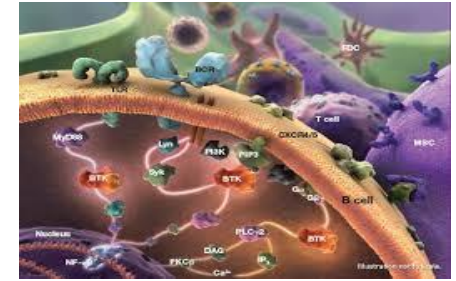
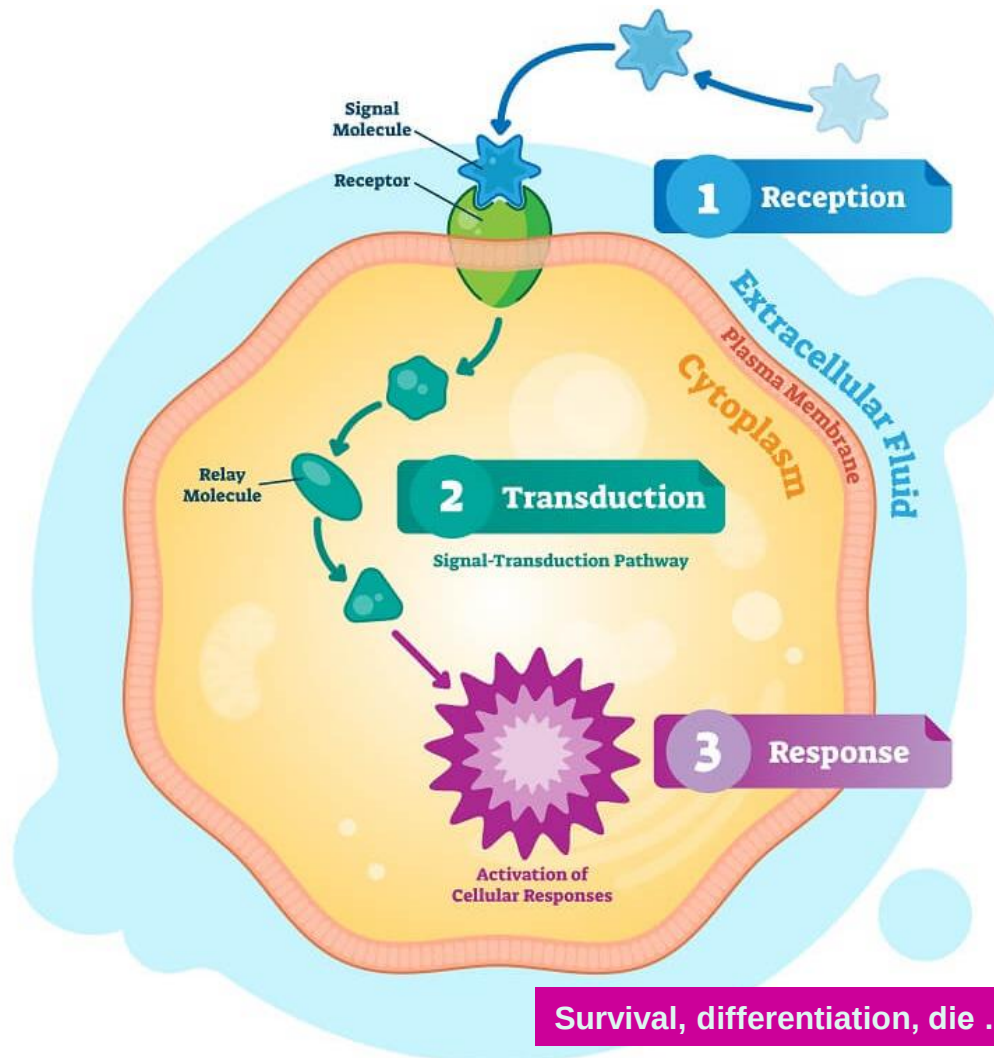
Cours I. Limon



Cours M. Glorian

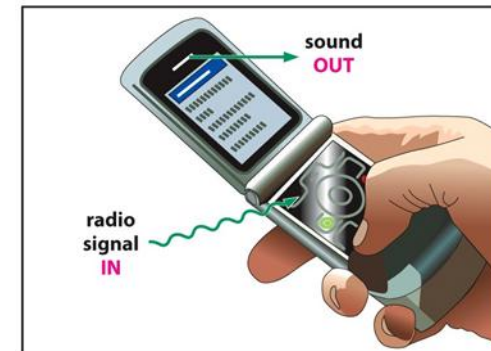
Indirect communication

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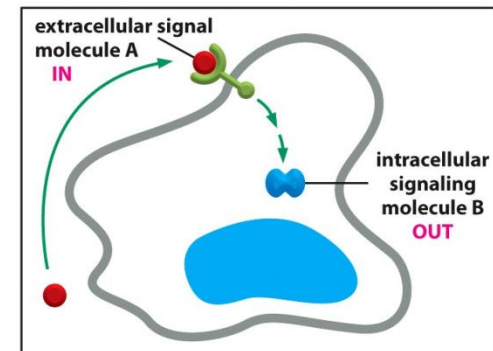
Signal transduction:

process whereby one type of signal is converted into another



(A)

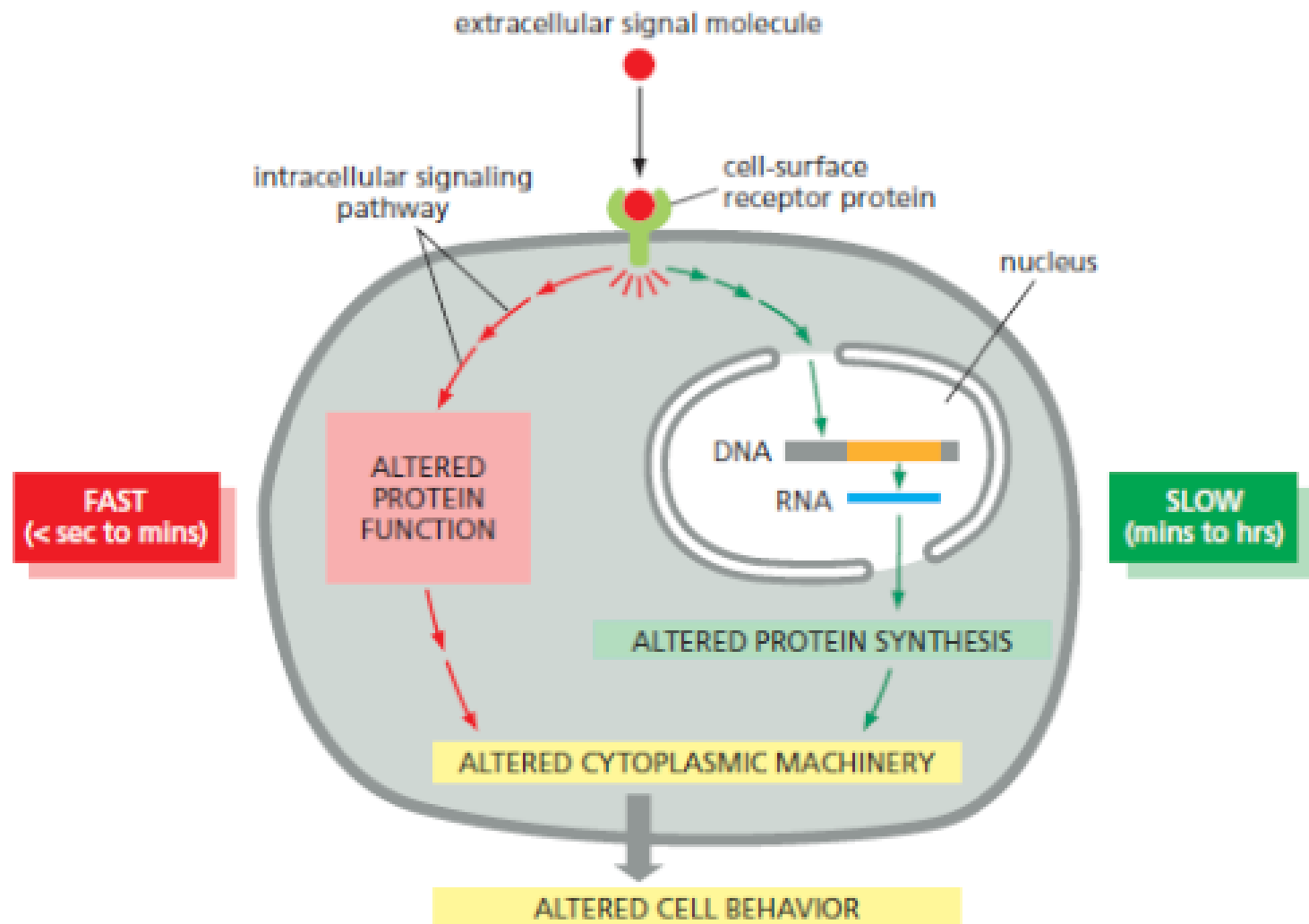
Figure 16-2 Essential Cell Biology 3/e (© Garland Science 2010)



(B)

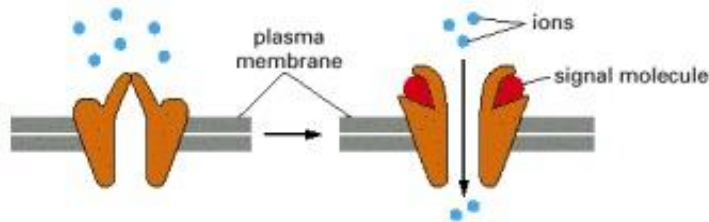
Indirect communication

A cell's response to a signal can be fast or slow:



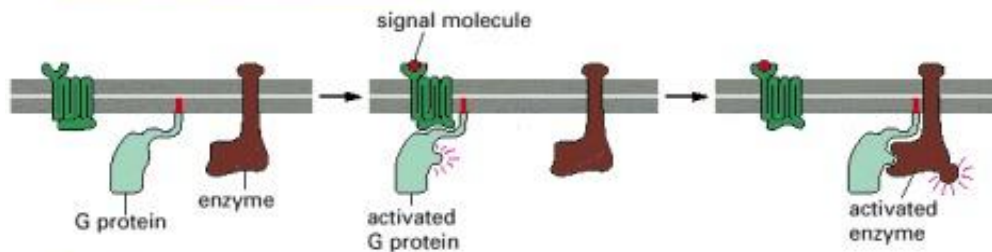
The Three Largest Classes of Cell-Surface Receptor Proteins

(A) ION-CHANNEL-LINKED RECEPTORS



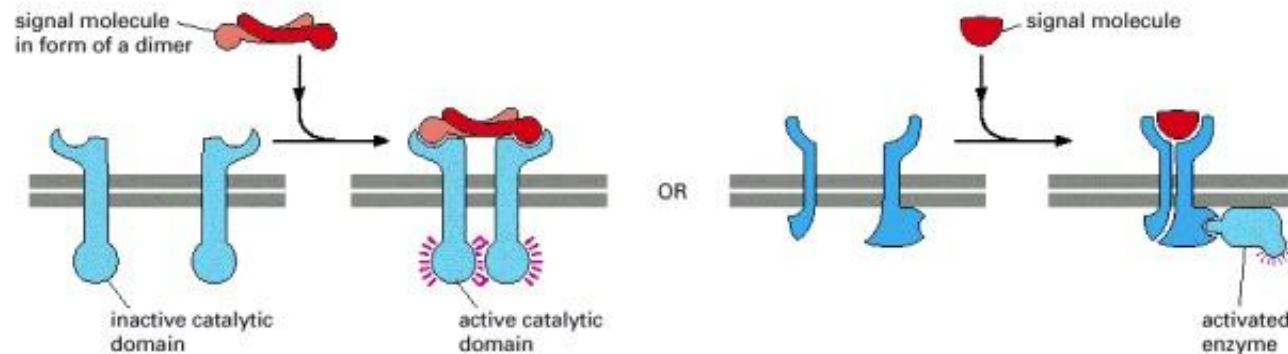
Ion-Channel-linked Rec.

(B) G-PROTEIN-LINKED RECEPTORS



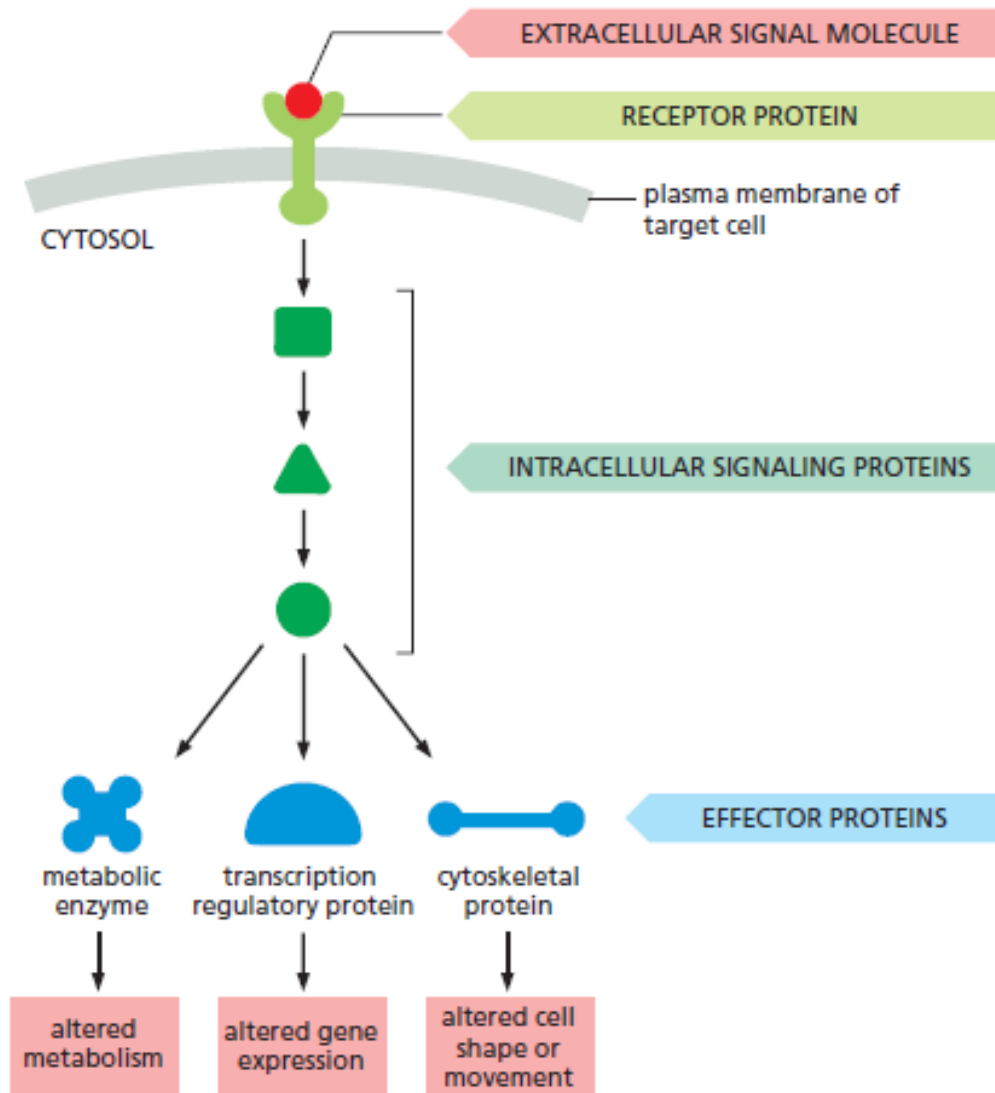
G-Protein-linked Rec.

(C) ENZYME-LINKED RECEPTORS



Enzyme-linked Rec.

Cell-surface receptors relay extracellular signals *via* intracellular signaling pathways



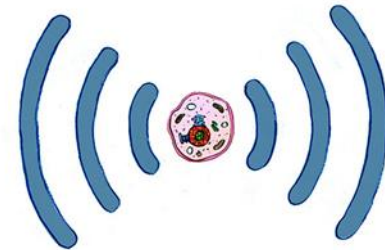
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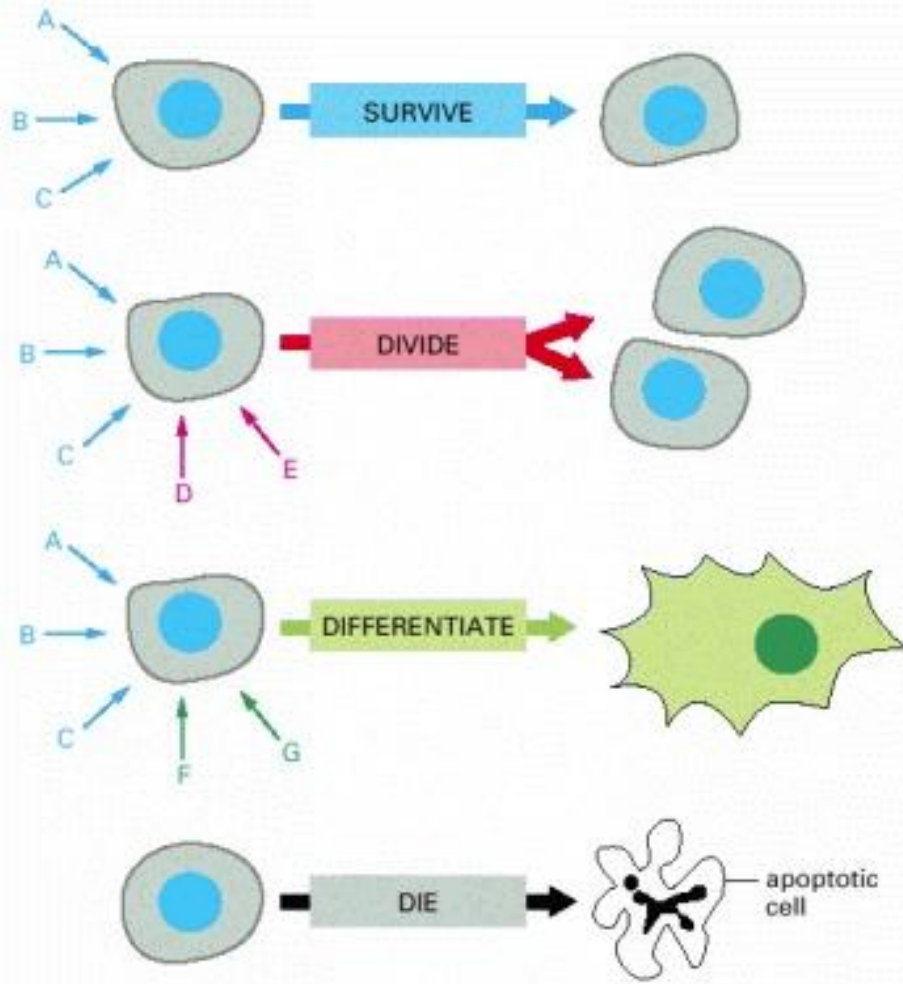
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Animal cell's dependence on multiple extracellular signals



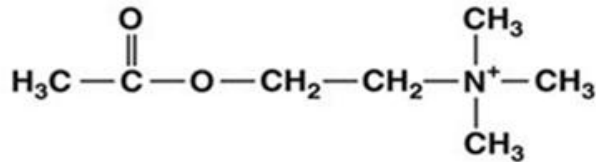
- Each cell type displays a **set of receptors** that enables it to respond to a corresponding **set of signal molecules** produced by other cells.
- Signal molecules work in combinations to regulate the **behavior of the cell**.
- As shown here, an individual cell requires multiple signals to **survive** (*blue arrows*) and additional signals to **divide** (*red arrow*) or **differentiate** (*green arrows*).
- If **deprived** of appropriate survival signals, a cell will undergo a form of cell suicide known as programmed cell death, or **apoptosis**.

Animal cell's dependence on multiple extracellular signals

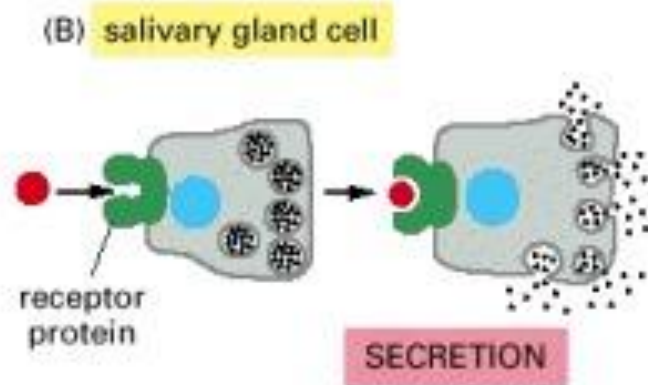
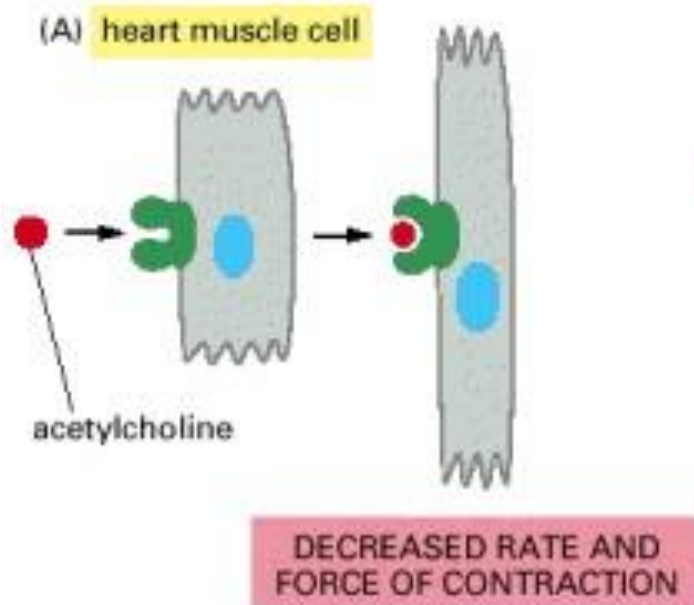
- A typical cell in a multicellular organism is exposed **to hundreds of different signals** in its environment. These signals can act in many **millions of combinations**.
- The cell must respond to this **babel** of signals **selectively**, according to its own **specific character**, which it has acquired through progressive cell specialization in the course of development.

Animal cell's dependence on multiple extracellular signals

(D) acetylcholine



Different cell types are specialized to respond to acetylcholine in different ways.



For these two cell types, acetylcholine binds to **similar receptor** proteins, but the intracellular signals produced **are interpreted differently** in cells specialized for different functions.

Animal cell's dependence on multiple extracellular signals

- A typical cell in a multicellular organism is exposed to **hundreds of different signals** in its environment. These signals can act in many **millions of combinations**.
- The cell must respond to this **babel** of signals **selectively**, according to its own **specific character**, which it has acquired through progressive cell specialization in the course of development.
- In principle, the **hundreds of signal molecules** that animals make can be used to create an **almost unlimited number of signaling combinations**.
- The use of these combinations to control cell behavior enables an animal to control its cells in **highly specific ways** by using **a limited diversity of signal molecules**.

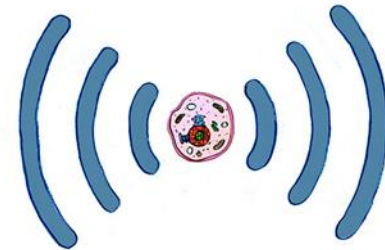
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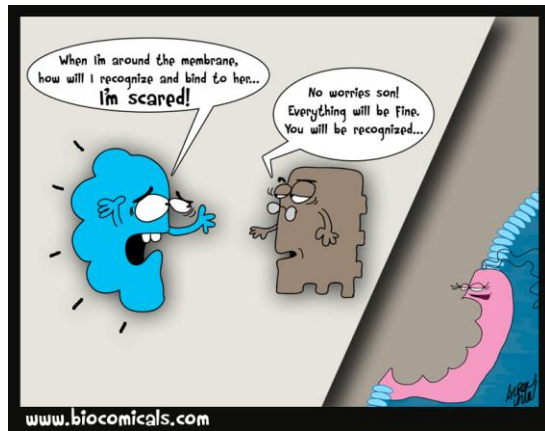
2.2.2/ *Proteins with enzymatic activity*

2.3- Second messengers

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3/ Cell signaling, a Complex but not complicated network

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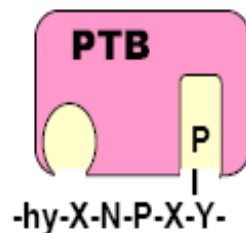
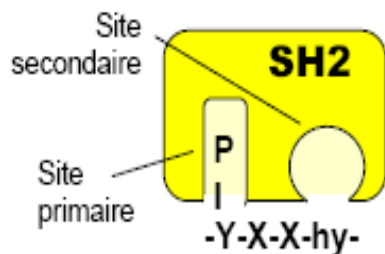
How does a signal remain strong, precise, and specific under the cellular noisy conditions?

- Ideally, an activated intracellular signaling molecule should interact only with the **appropriate downstream targets**, and, likewise, the targets should only be activated by the **appropriate upstream signal**.
- Intracellular signaling molecules share the cytoplasm with a **crowd** of closely related signaling molecules that control a **diverse array of cellular processes**.

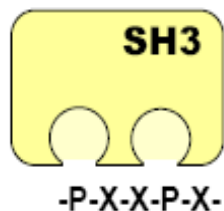
The first line of defense comes from the high affinity and specificity of the interactions between signaling molecules and their correct partners.

There are many types of interaction domains in signaling proteins

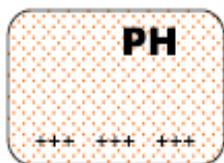
Interaction domains enable the signaling protein they are part of to **dock** on the **membrane** and interact with other similarly recruited **signaling proteins**.



Src homology 2 (**SH2**) domains and phosphotyrosine-binding domains (**PTB**) bind to phosphorylated tyrosines in a **particular peptide sequence** on activated receptors or intracellular signaling proteins.

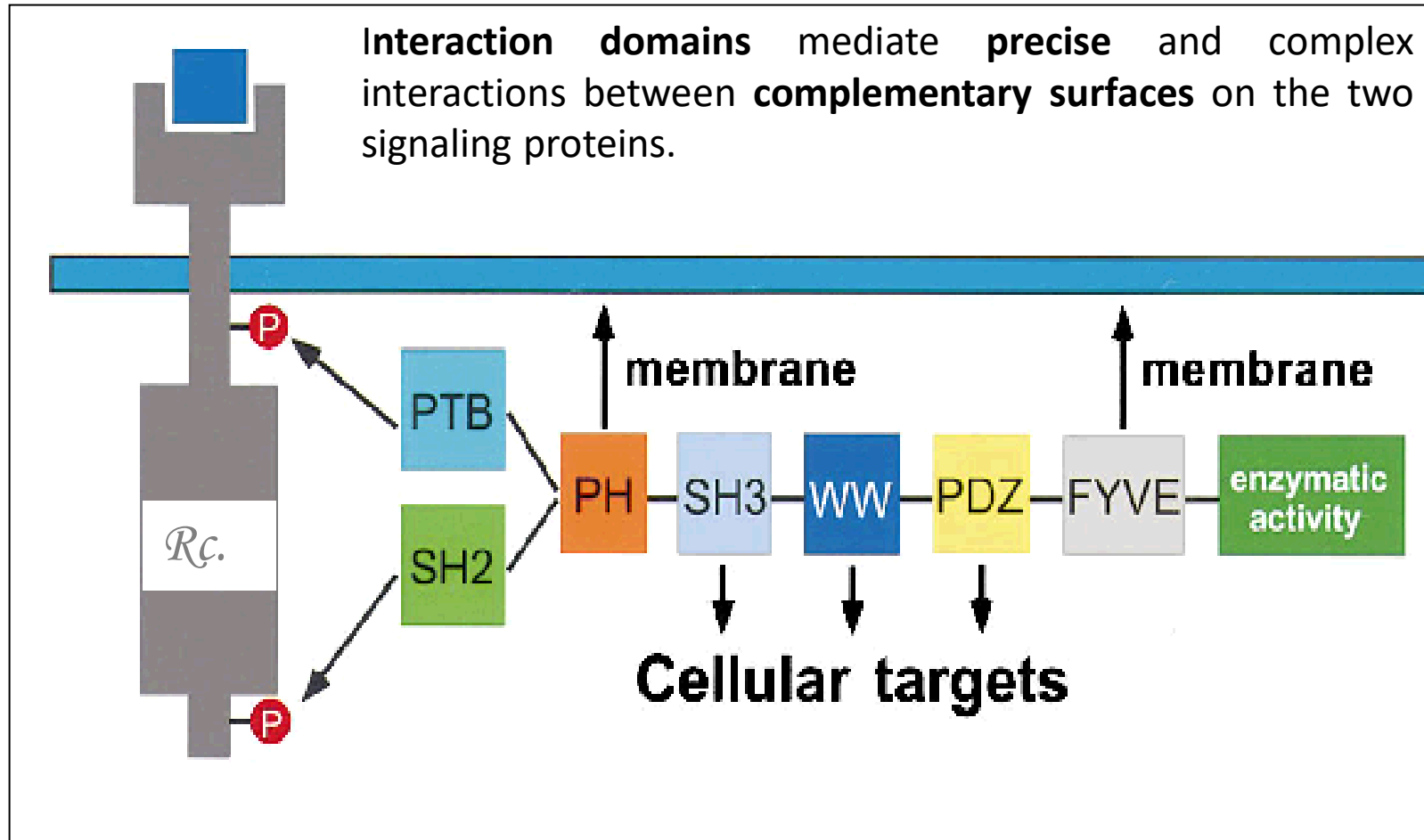


Src homology 3 (**SH3**) domains bind to short, **proline-rich** amino acid sequences.



Phospholipides

Some pleckstrin homology (**PH**) domains bind to the **charged** head groups of specific **phosphoinositides** that are produced in the plasma membrane in response to an extracellular signal.



Signaling proteins

Adaptor proteins

Proteins with enzymatic activity
(kinase, lipase, GTPase...)

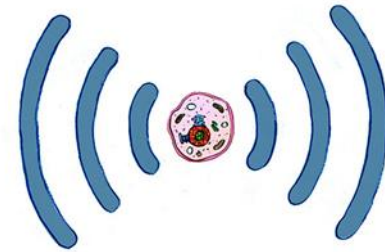
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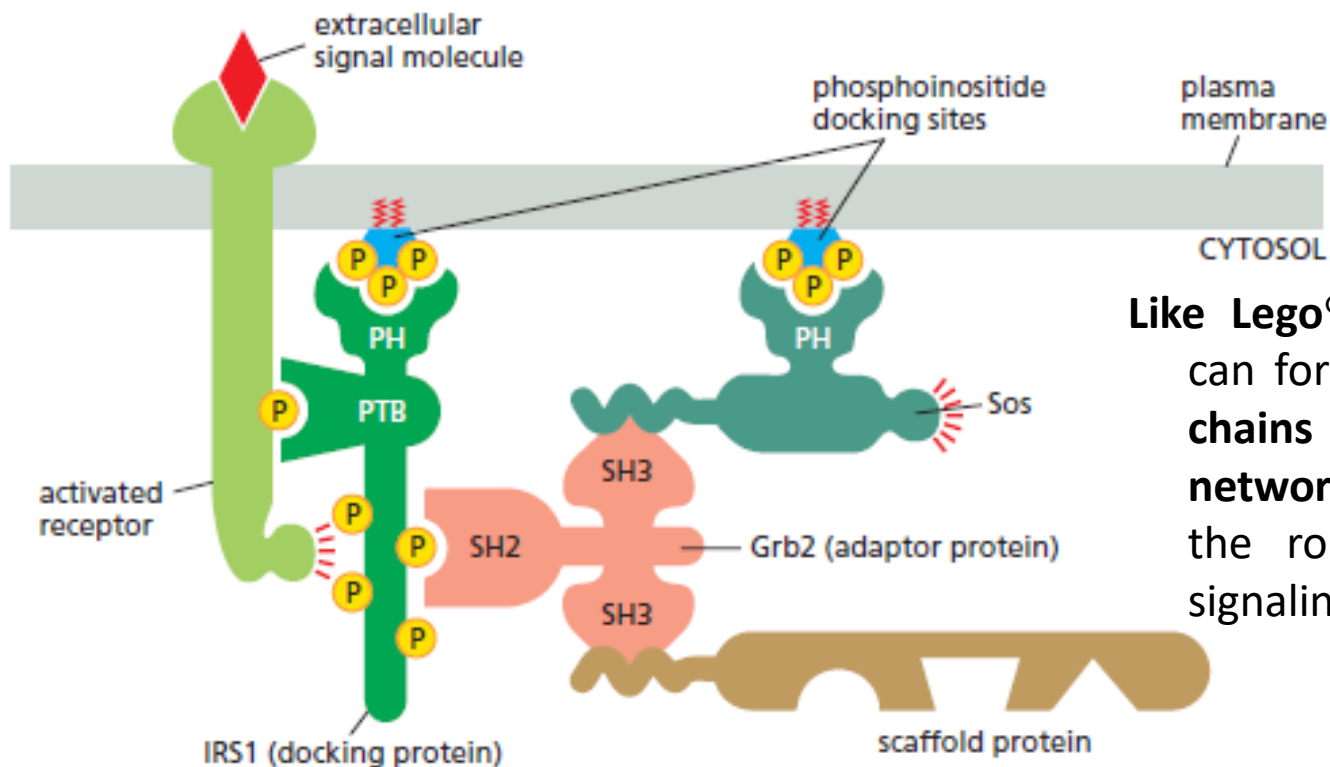
2.3- Second messengers

2.4- Complexes enhance the speed, efficiency, and Specificity of Intracellular Signaling

3/ Cell signaling, a Complex but not complicated network

III/ cell signaling and pathologies

- Some signaling proteins consist solely of **two or more** interaction domains and function only as **adaptors** to link two other proteins together in a signaling pathway.
- Interaction domains enable signaling proteins to bind to one another in **multiple specific combinations**.
- **Conformational modifications** induce signaling partner activation.



Like Lego® bricks, the proteins can form **linear** or **branching chains** or **three dimensional networks**, which determine the route followed by the signaling pathway.

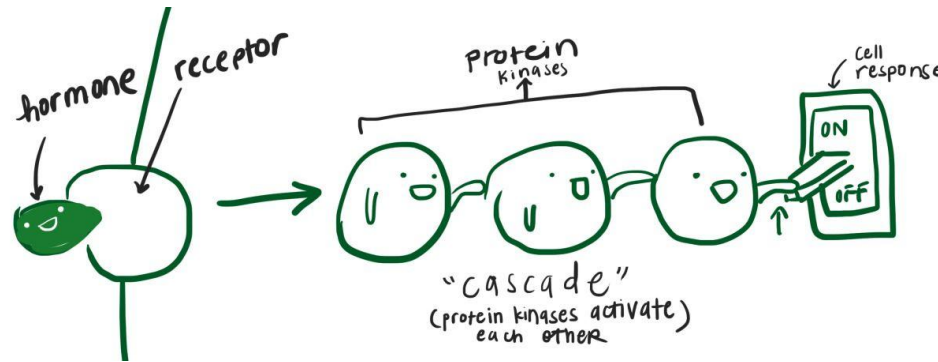
Some Intracellular Signaling Proteins Act as Molecular Switches

- On receipt of a signal they switch from an inactive to an active state, until another process switches them off. **The switching off is just as important as the switching on.**
- If a signaling pathway is to **recover** after transmitting a signal so that it can be ready to transmit another, every activated molecule in the pathway must be **returned** to its original **inactivated state**.
- The molecular switches fall into two main classes, although in both cases it is the **gain** or **loss** of **phosphate groups** that determines whether the protein is active or inactive.
- The largest class consists of proteins that are activated or inactivated by **phosphorylation**. For these proteins, the switch is thrown in one direction by a **protein kinase**, which adds one or more phosphate groups to the signaling protein, and in the other direction by a **protein phosphatase**, which removes the phosphate groups from the protein.

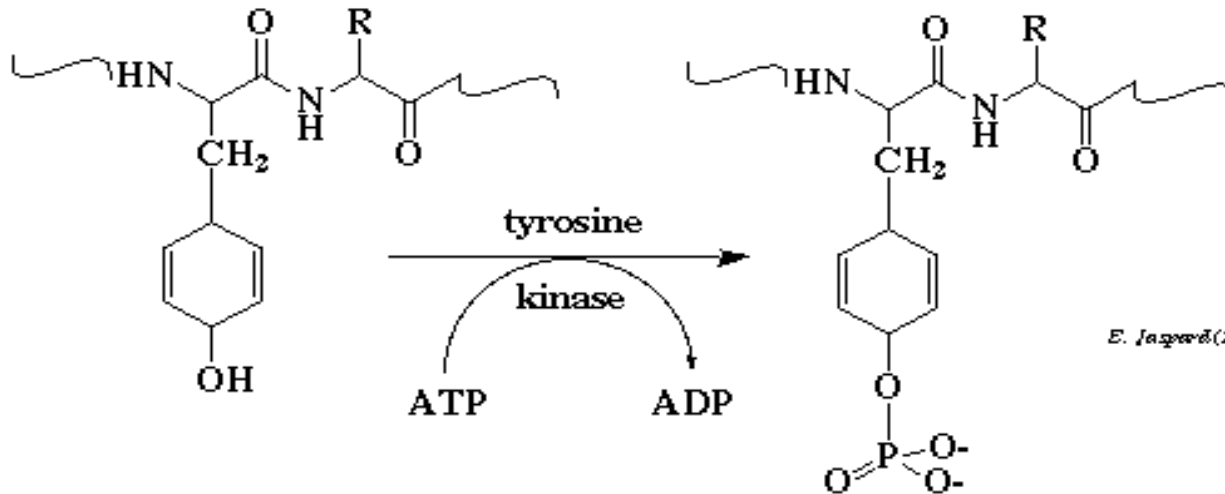
Signaling proteins with enzymatic activity

Two types of intracellular signaling proteins that act as molecular switches

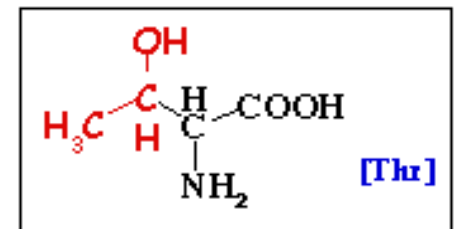
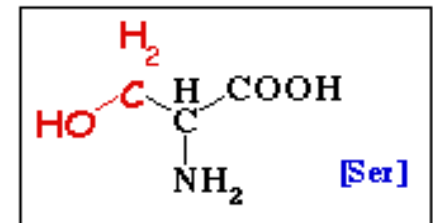
Kinases



Protein kinases **attach phosphate** to the hydroxyl group of **specific amino acids** on the target protein.



E. Jasperd (2004)

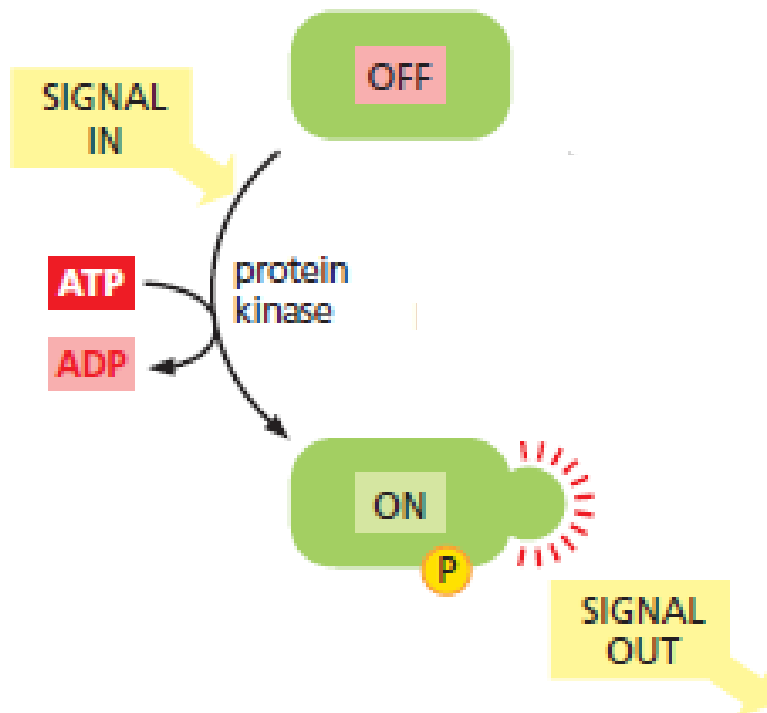


Signaling proteins with enzymatic activity

Two types of intracellular signaling proteins that act as molecular switches

Kinases

- A protein Kinase **covalently adds a phosphate** from ATP to the signaling protein and protein phosphatase **removes the phosphate**.
- Phosphorylation can **activates** (ON) or **inactivates** (OFF) the target protein.



Signaling proteins with enzymatic activity

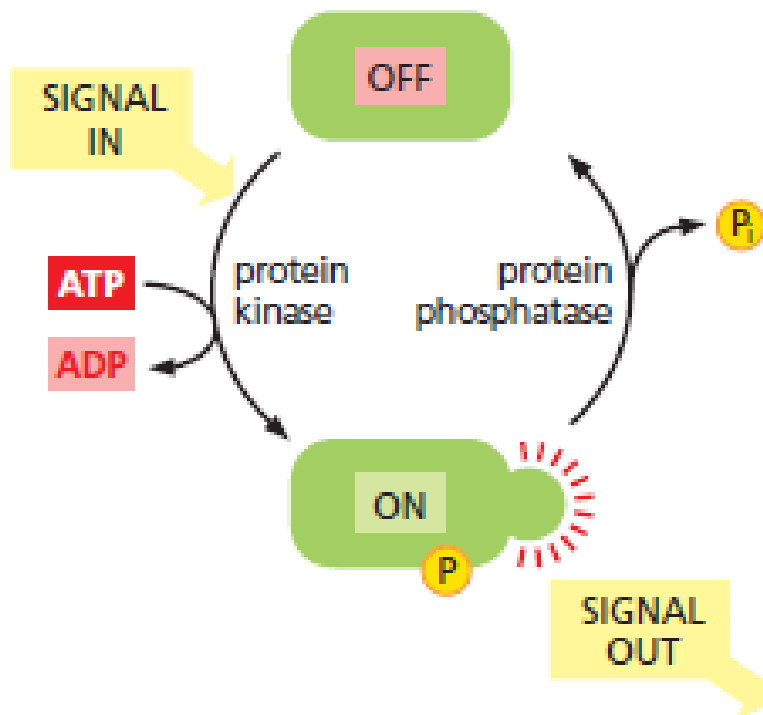
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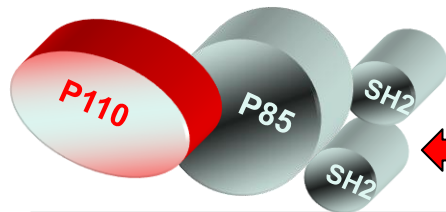
Kinases/Phosphatases

- The activity of any protein regulated by phosphorylation depends on the **balance** between the **activities of the kinases** that phosphorylate it and of **the phosphatases** that dephosphorylate it.



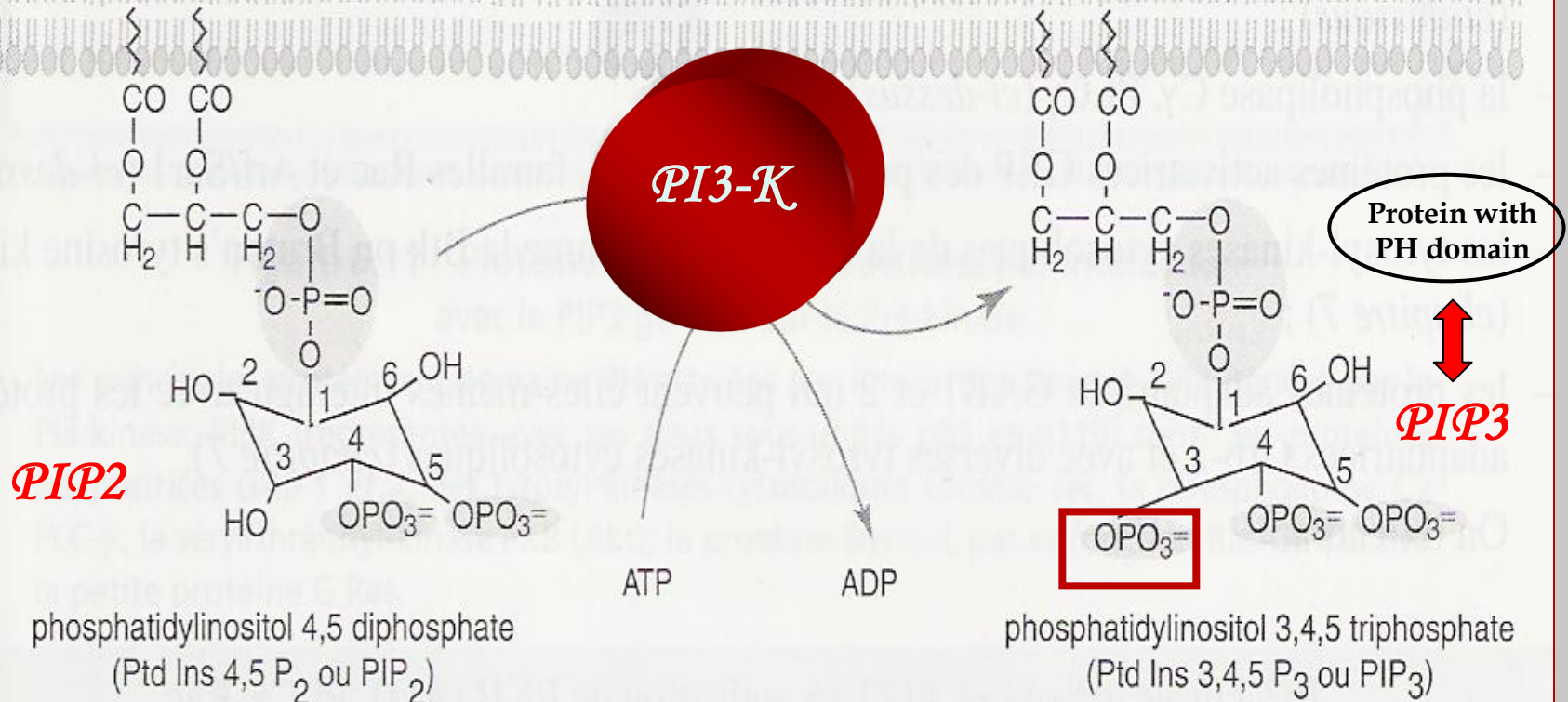
About **30–50%** of human proteins contain covalently attached phosphate, Genome sequencing reveals that about **2% of our genes encode protein kinases**, and it is thought that hundreds of distinct types of protein kinases are present in a typical

Exemple of Kinase: Phosphatidyl-Inositol3-Kinase (PI3-K)



P-Y-Protein
(Adaptor protein or Rc)

PI3-K: heterodimer



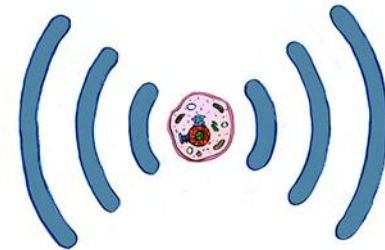
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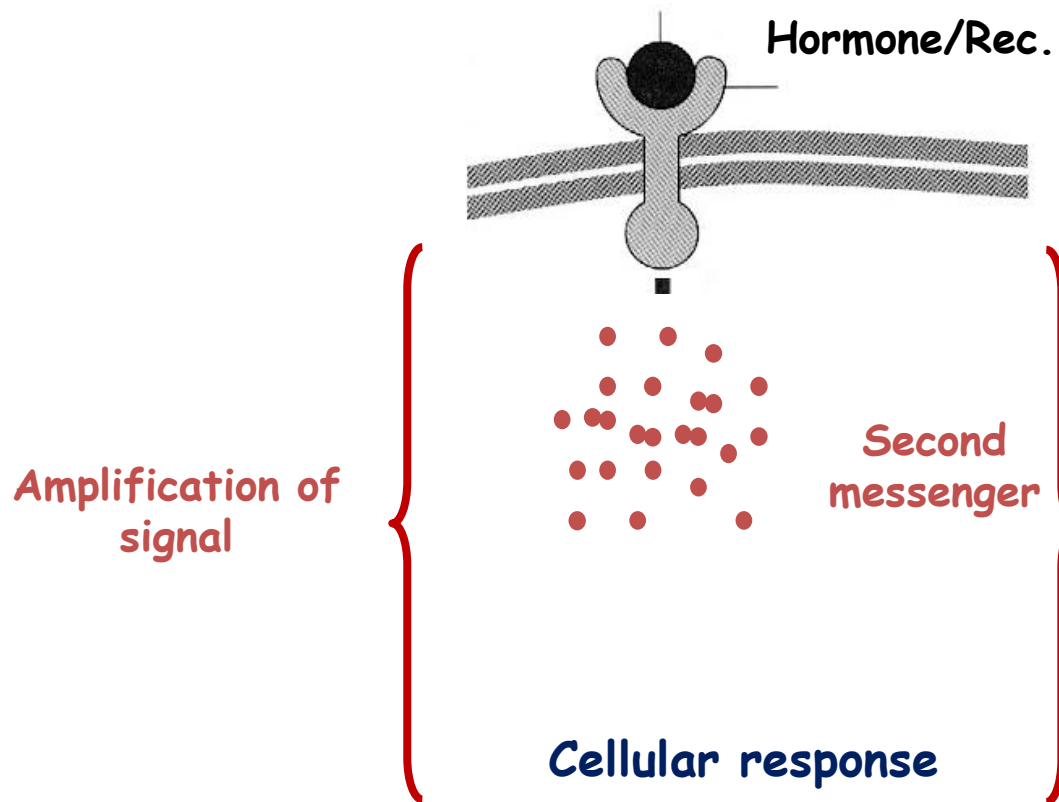
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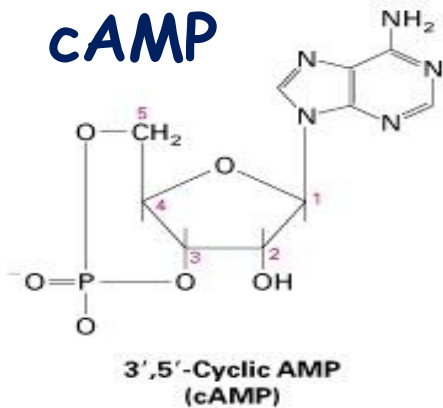
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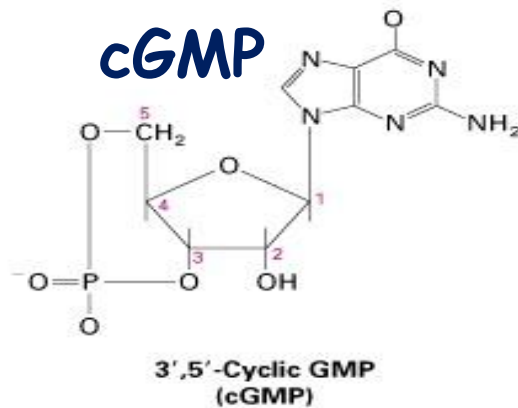
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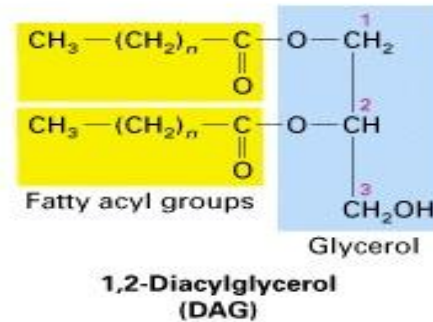
cAMP



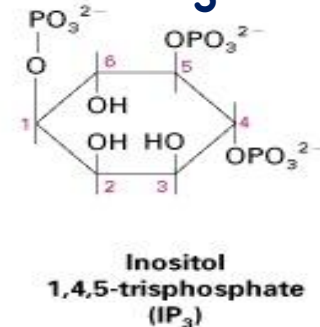
cGMP



DAG



IP₃



Second messengers

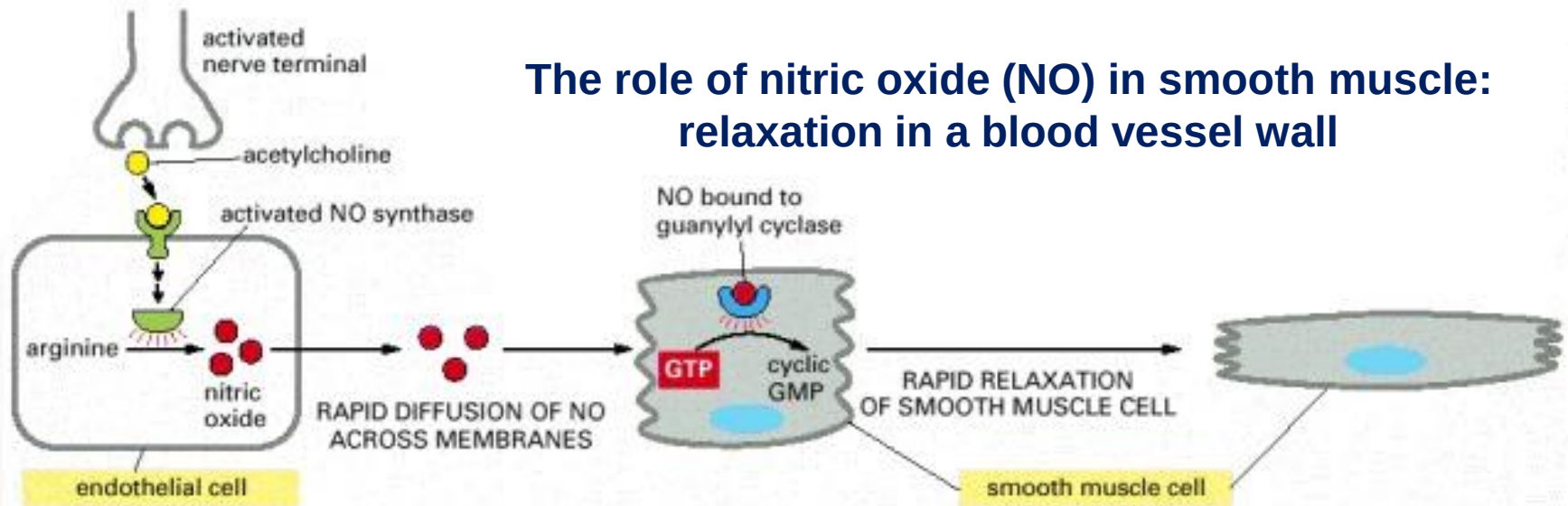
The small intracellular signaling molecules are called **small intracellular mediators**, or **second messengers** (the “first messengers” being the extracellular signals).

They are generated in large numbers in response to receptor activation and **rapidly diffuse** away from their source, **broadcasting the signal** to other parts of the cell.

Some, such as *cyclic AMP* (**AMPC**) and **Ca²⁺**, are water-soluble and diffuse in the cytosol, while others, such as **diacylglycerol**, are lipid-soluble and diffuse in the plane of the plasma membrane or across it as **Nitric Oxide (NO)**.

Nitric Oxide Gas Signals by Binding Directly to an Enzyme Inside the Target Cell

The role of nitric oxide (NO) in smooth muscle: relaxation in a blood vessel wall



- Acetylcholine released by nerve terminals in the blood vessel wall **activates NO synthase** in **endothelial cells** lining the blood vessel, causing the endothelial cells to produce NO. The **NO diffuses** out of the endothelial cells and into the underlying smooth muscle cells, where it binds to and activates guanylyl cyclase to **produce cyclic GMP**. The cyclic GMP triggers a response that causes the **smooth muscle cells to relax**, enhancing blood flow through the blood vessel.
- The NO released by **autonomic nerves** in the **penis** causes the local blood vessel dilation that is responsible for penile **erection**.
- NO is also produced as a **local mediator** by **activated macrophages** and neutrophils to help them to **kill invading microorganisms**.

Second messengers

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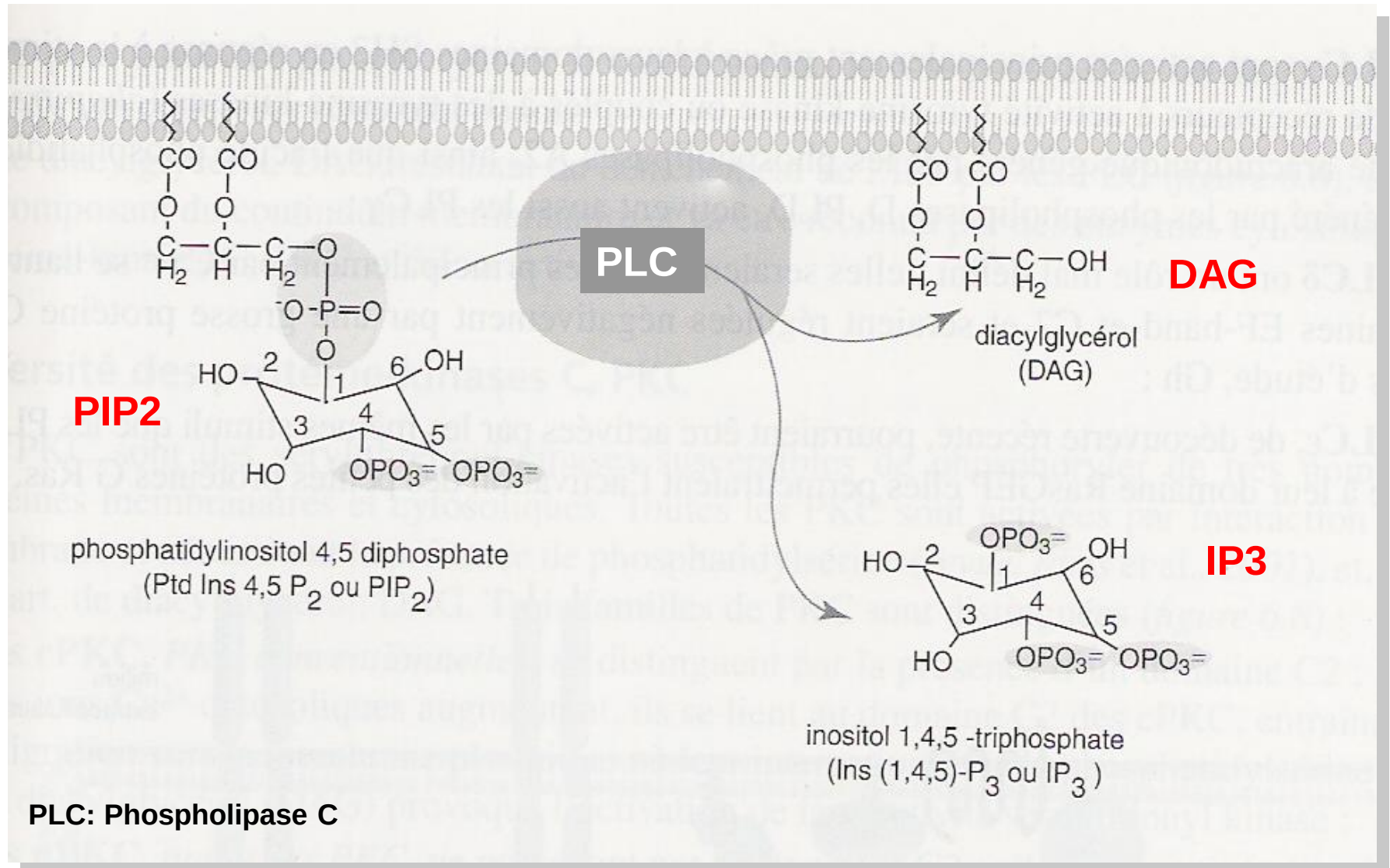
They are generated in large numbers in response to receptor activation and **rapidly diffuse** away from their source, **broadcasting the signal** to other parts of the cell.

Some, such as *cyclic AMP* (**AMPC**) and **Ca²⁺**, are water-soluble and diffuse in the cytosol, while others, such as **diacylglycerol**, are lipid-soluble and diffuse in the plane of the plasma membrane or across it as **Nitric Oxide (NO)**.

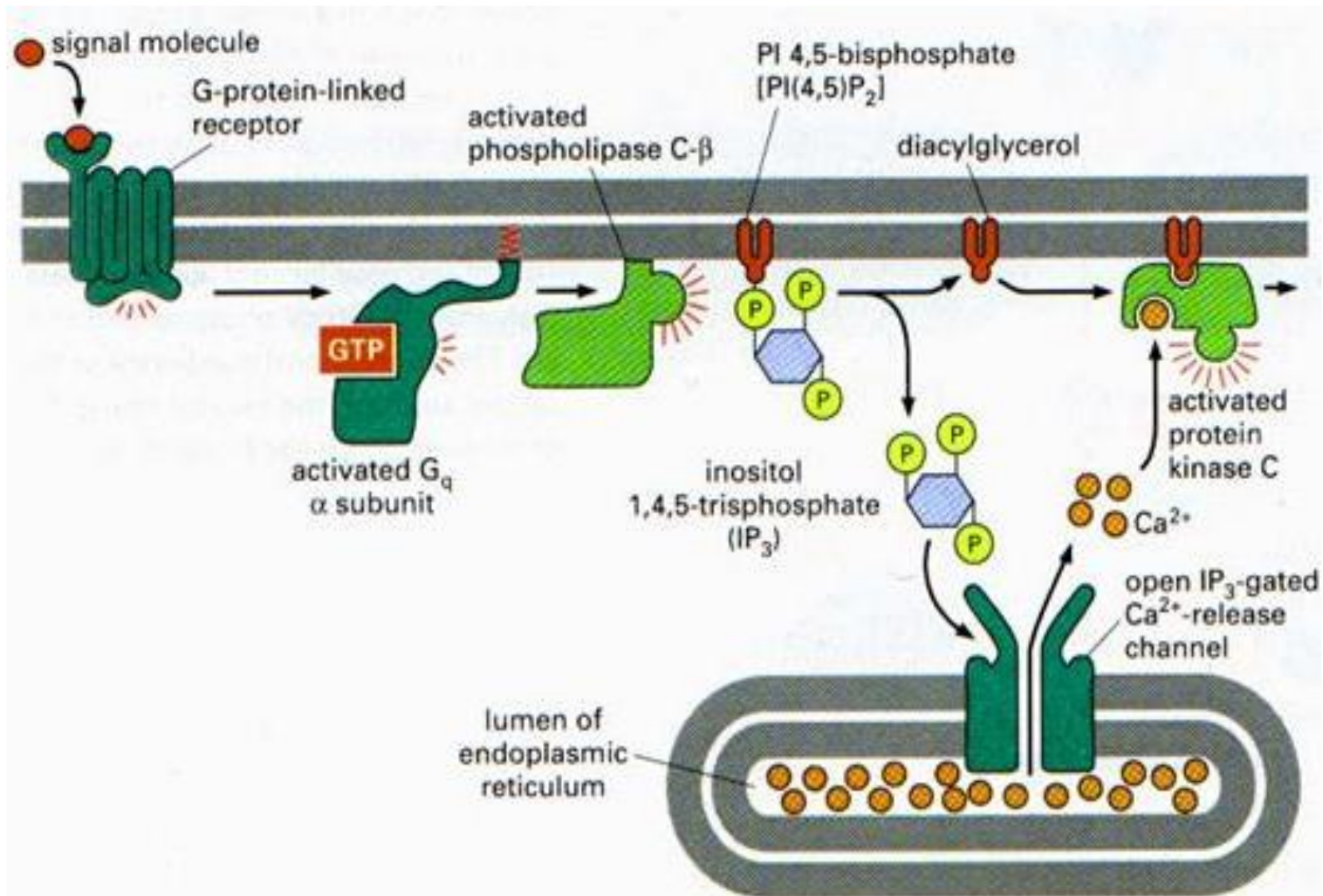
In either case, they pass the signal on by **binding** to and **altering the behavior** of selected signaling proteins **or target proteins**.

Not all molecular switches in signaling systems depend on **phosphorylation**. Some signaling proteins are switched on or off by the binding of **adaptor protein** or a **second messenger** (cyclic AMP, Ca²⁺...), or by modifications other than phosphorylation or dephosphorylation, such as **ubiquitylation** or **GTP binding**.

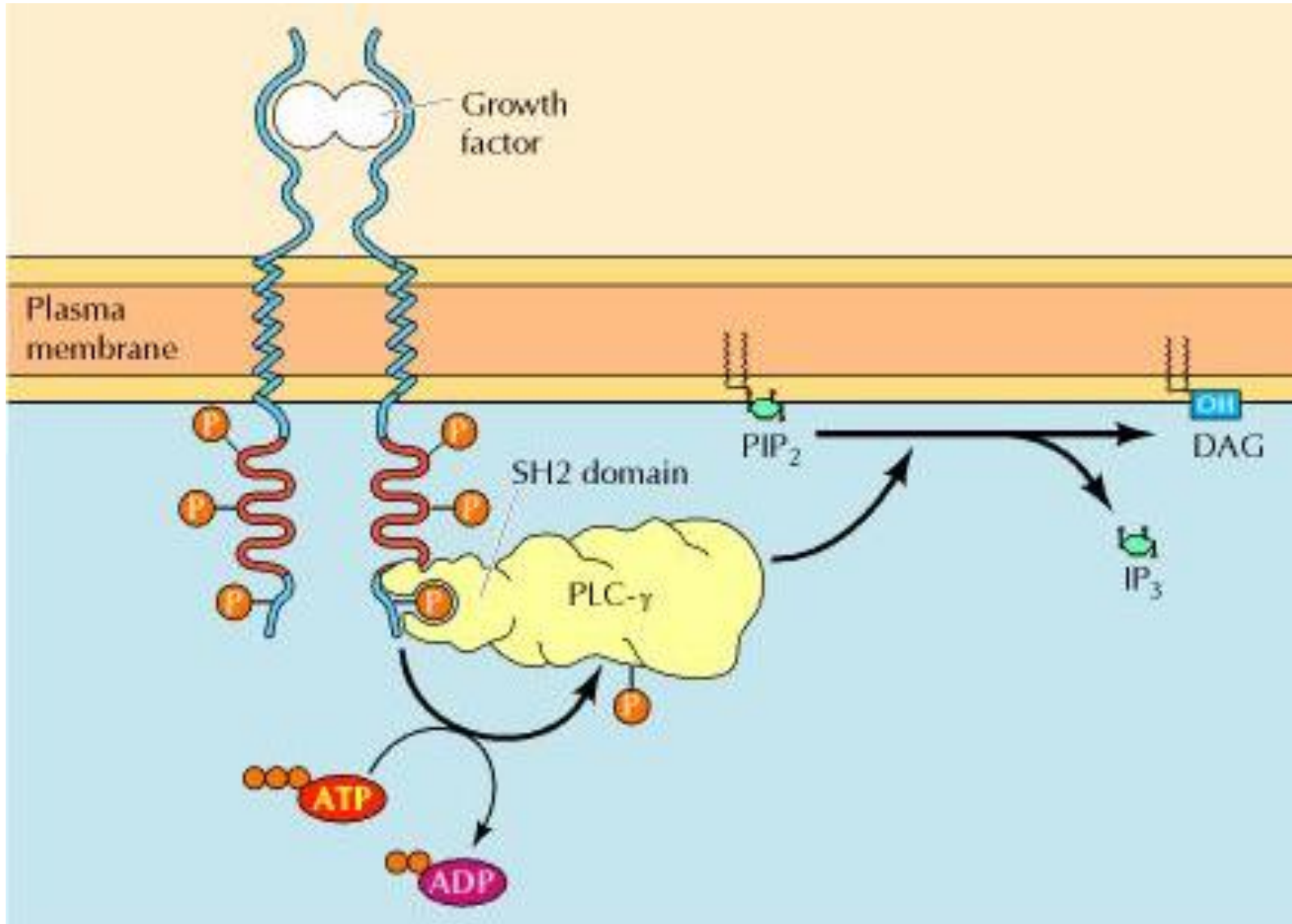
DAG and IP3 Production



IP3 and DAG Pathway

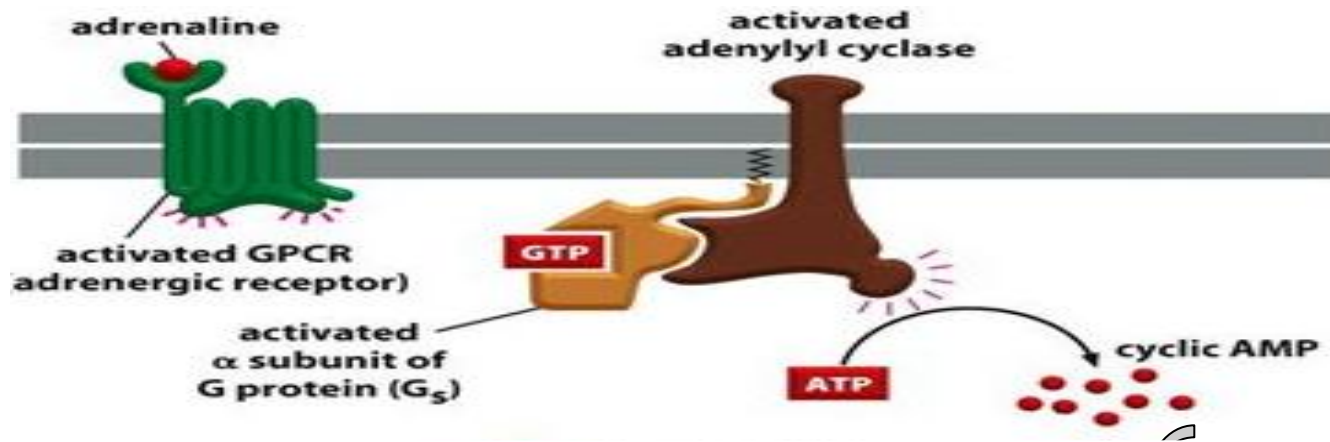


IP3 and DAG Pathway



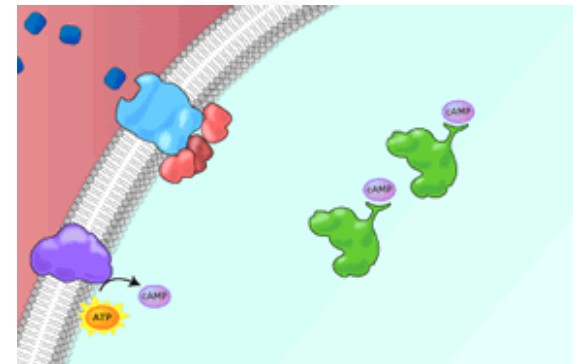
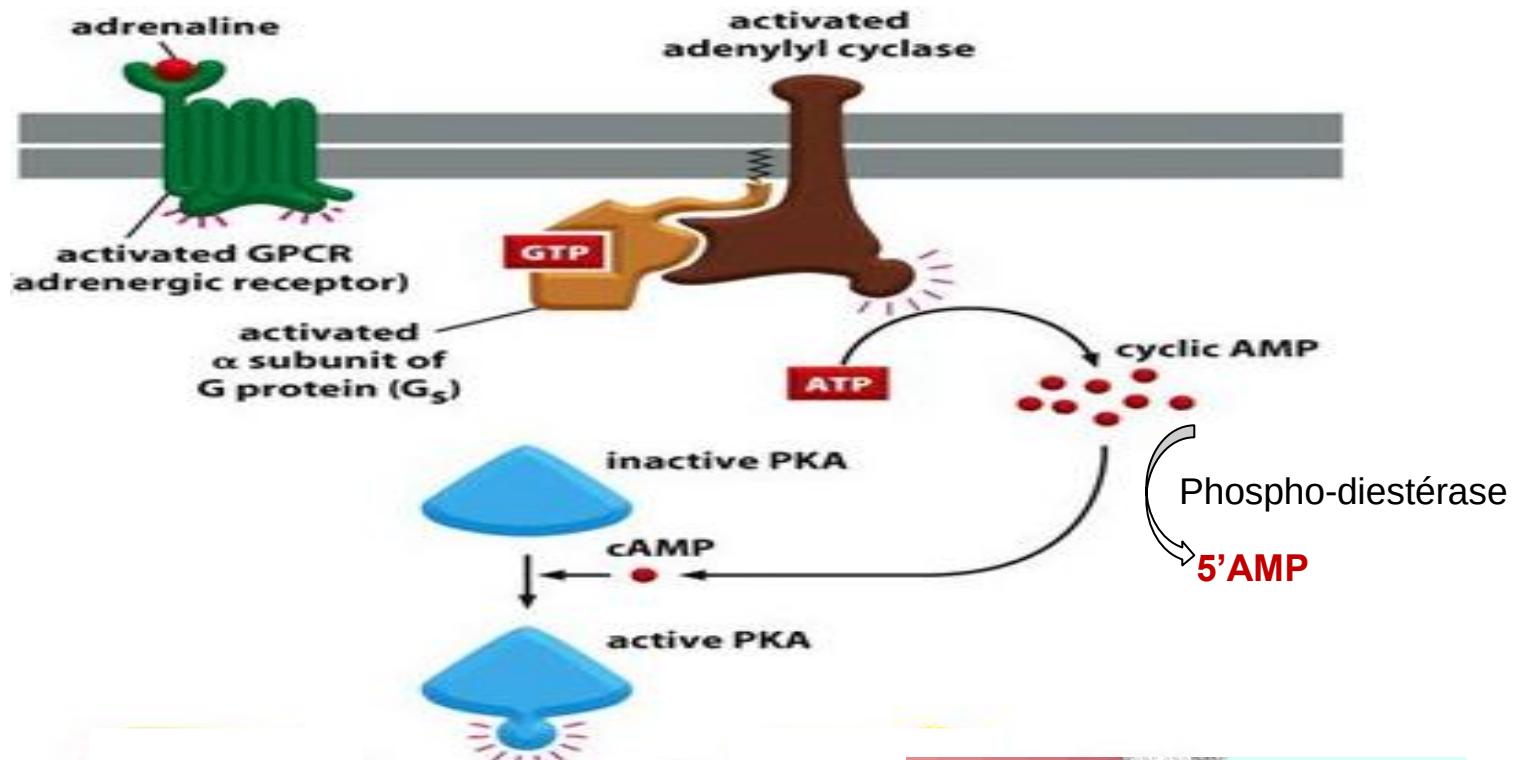
Cyclic AMP (cAMP) pathway

Example: adrenaline stimulates glycogen breakdown in skeletal muscle (Fast response).



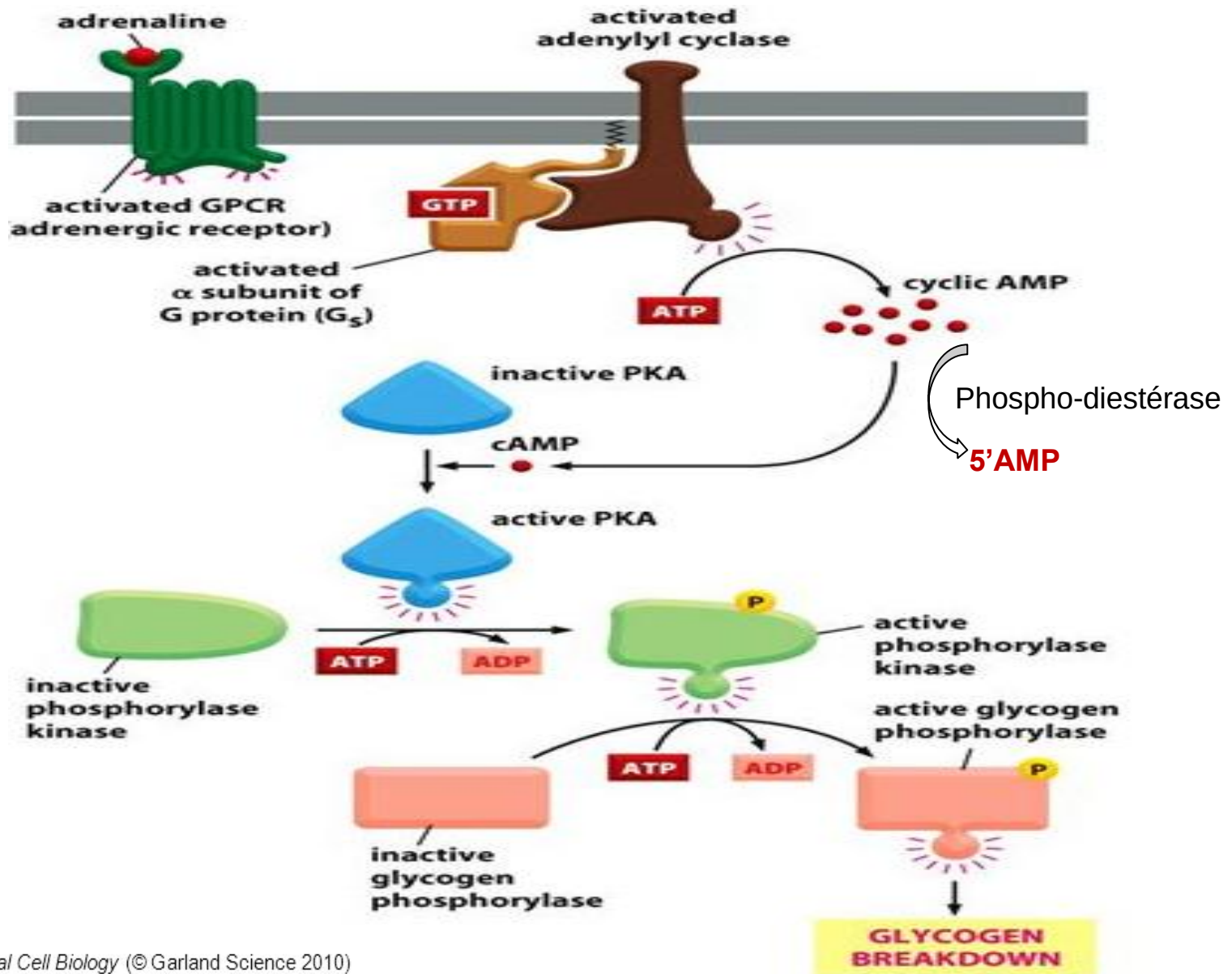
Cyclic AMP (cAMP) pathway

Exemple: adrenaline stimulates glycogen breakdown in skeletal muscle (Fast response).

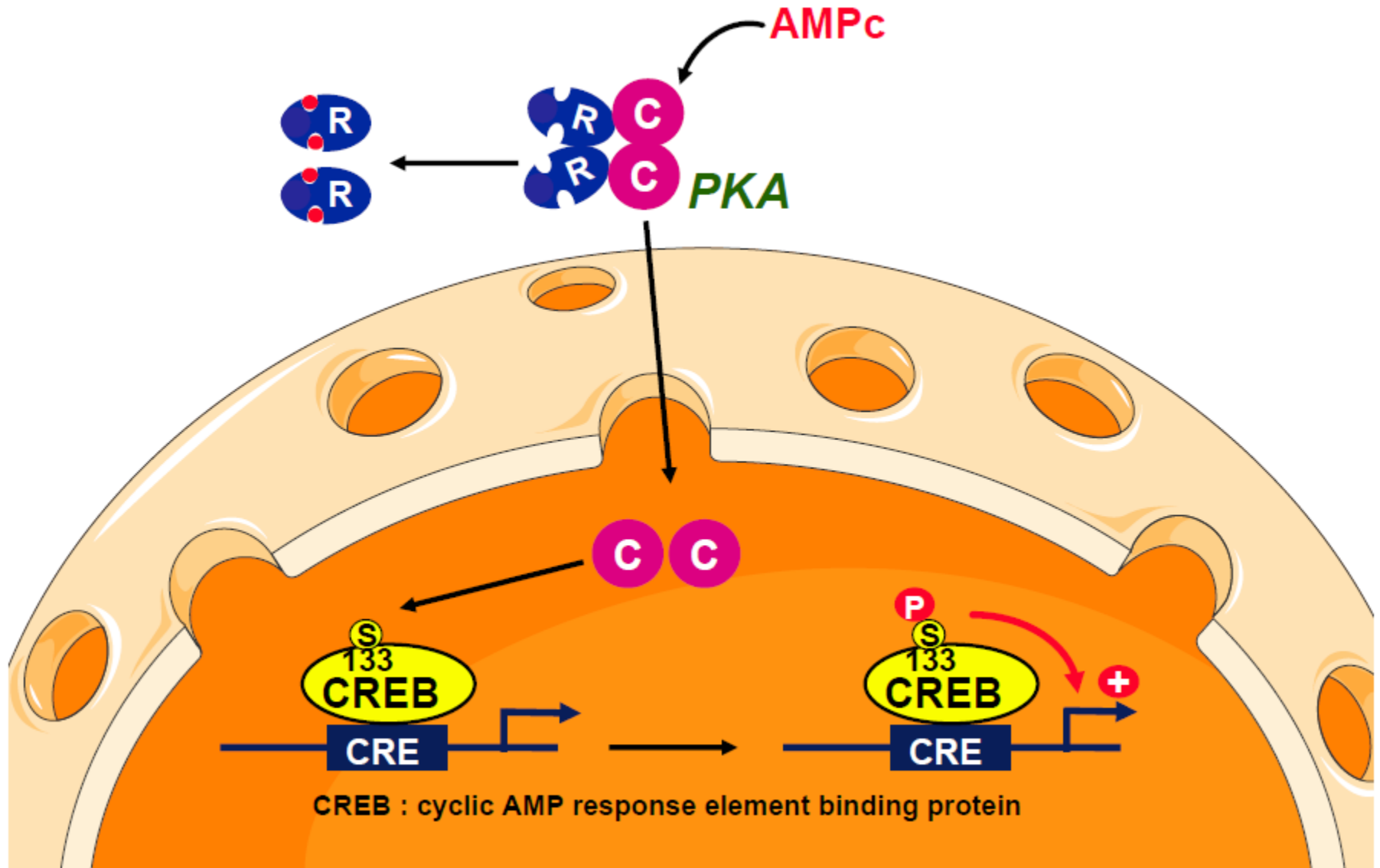


Cyclic AMP (cAMP) pathway

Exemple: adrenaline stimulates glycogen breakdown in skeletal muscle (Fast response).



Slow response to cAMP production



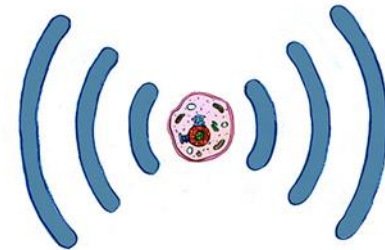
General Principles of cell signaling

I/ Introduction to cellular communication

II/ Actors of cell signal transduction

1/ Forms of chemical signaling

2/ Actors of signaling pathways



2.1 How does a signal remain strong, precise, and specific ?

2.2- Signaling proteins

2.2.1/ *adaptor proteins*

2.2.2/ *Proteins with enzymatic activity*

2.3- Second messengers

2.4- Complexes enhance the speed, efficiency, and Specificity of Intracellular Signaling

3/ Cell signaling, a Complex but not complicated network

III/ cell signaling and pathologies

Intracellular Signaling Complexes Enhance the Speed, Efficiency, and Specificity of the Response

It is not clear how an individual cell manages to display specific responses to so many different extracellular signals, many of which bind to the **same class of receptor** and activate many of the **same signaling pathways**.

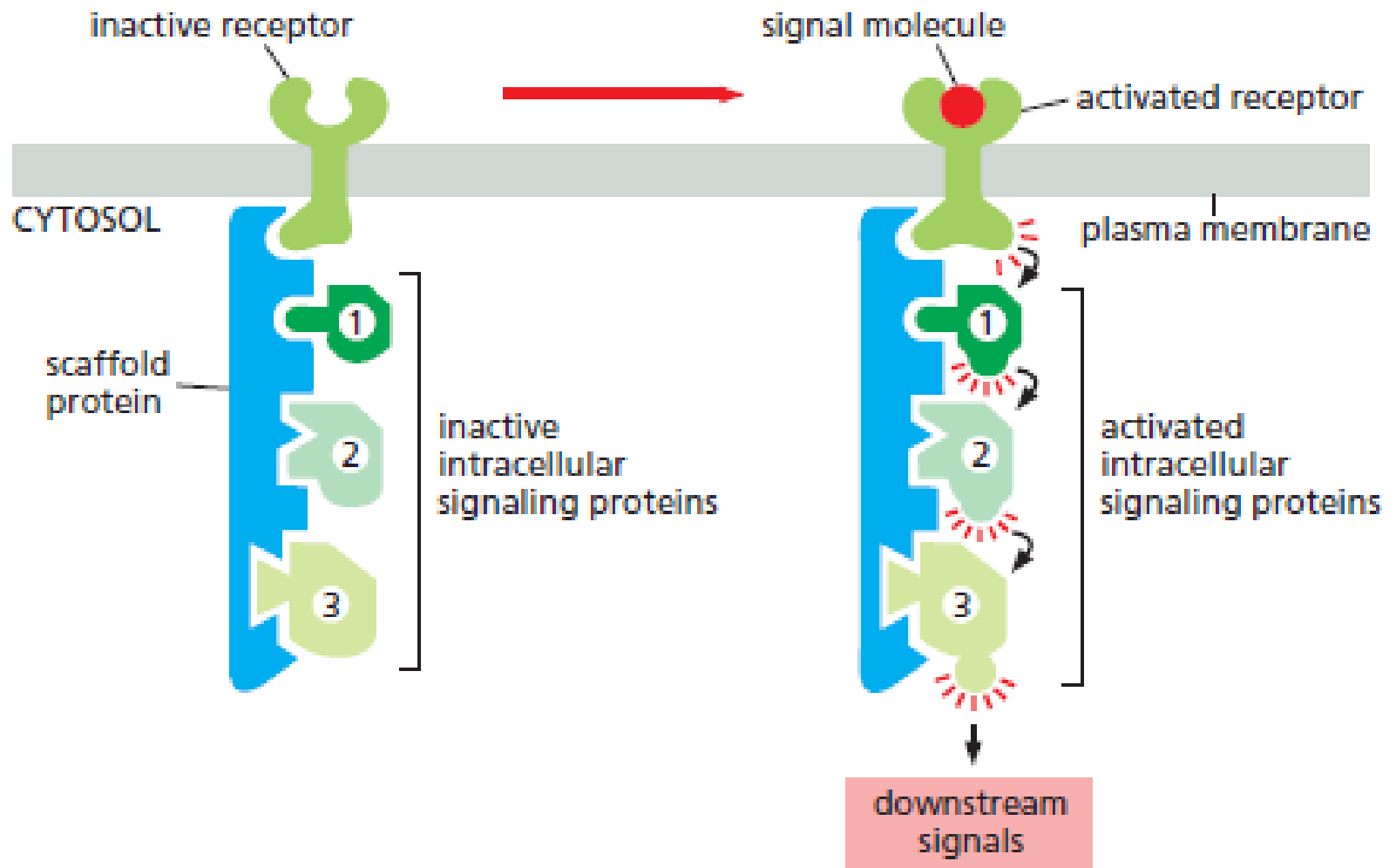
One strategy that the cell uses to achieve specificity involves **scaffold proteins** which organize groups of interacting signaling proteins into *signaling complexes*.

Because the scaffold guides the interactions between the successive components in such a complex, the signal can be relayed **with precision, speed, and efficiency**; moreover, unwanted cross-talk between signaling pathways is avoided.

In order to amplify a signal, however, and spread it to other parts of the cell, at least some of the components in most signaling pathways are likely to be **freely diffusible**.

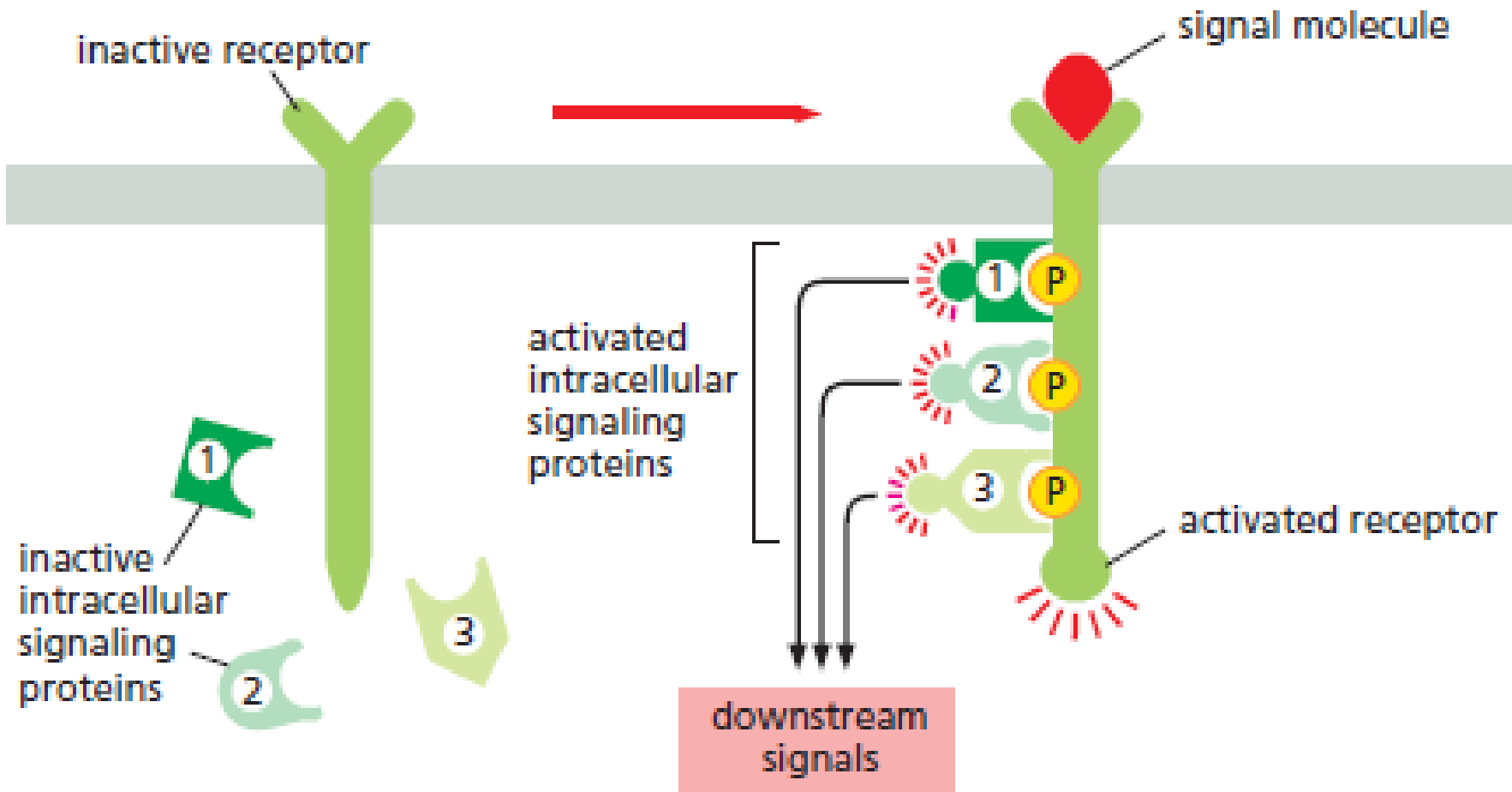
Intracellular Signaling Complexes Enhance the Speed, Efficiency, and Specificity of the Response

(A) PREFORMED SIGNALING COMPLEX ON A SCAFFOLD PROTEIN



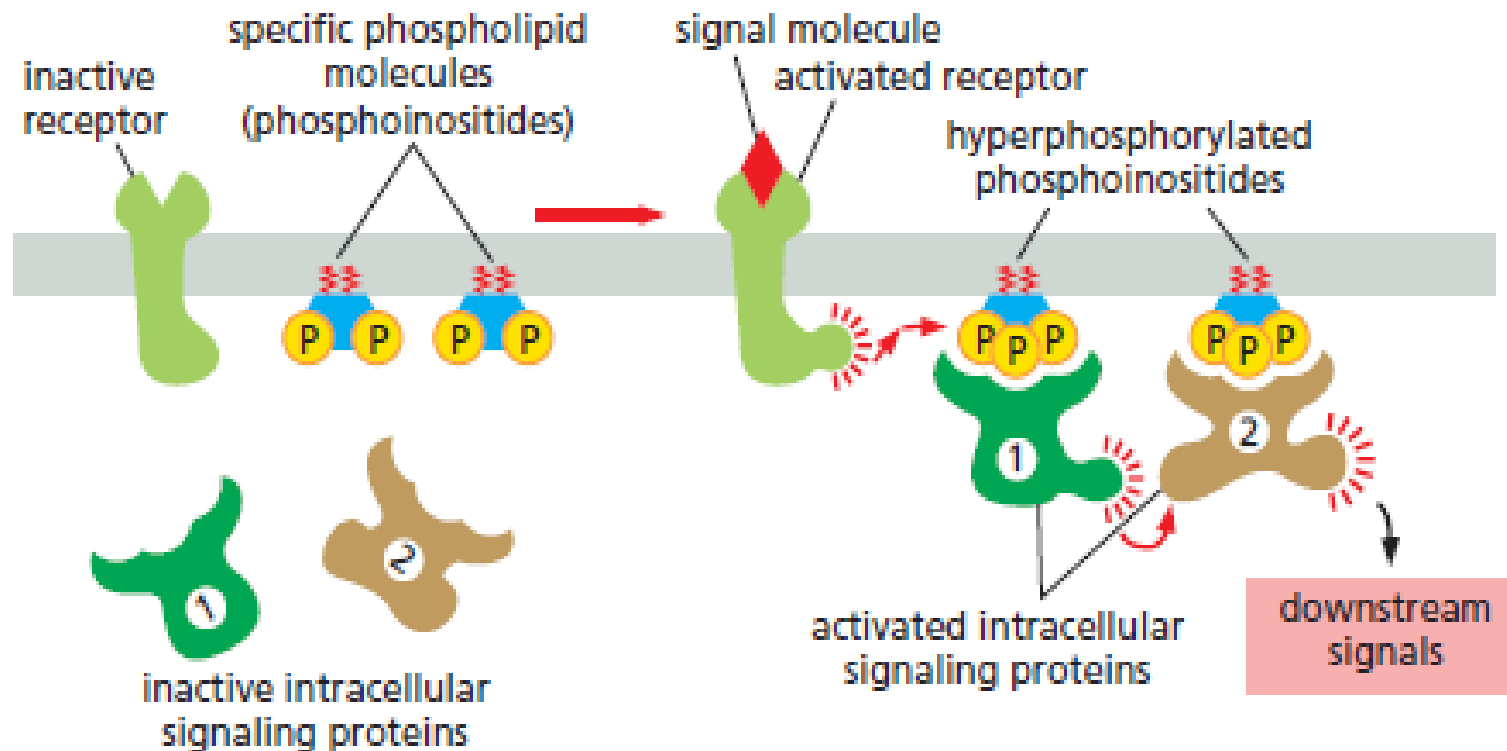
Intracellular Signaling Complexes Enhance the Speed, Efficiency, and Specificity of the Response

(B) ASSEMBLY OF SIGNALING COMPLEX ON AN ACTIVATED RECEPTOR



Intracellular Signaling Complexes Enhance the Speed, Efficiency, and Specificity of the Response

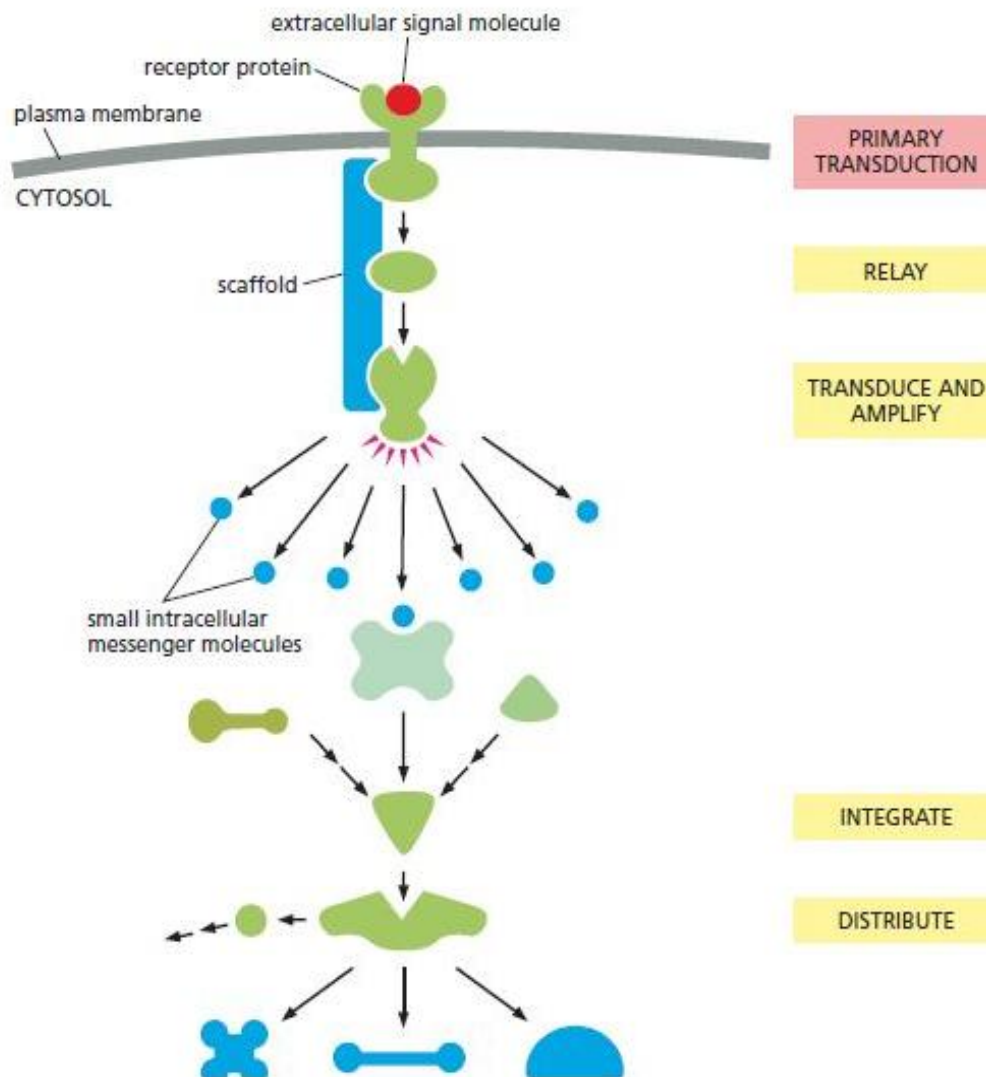
(C) ASSEMBLY OF SIGNALING COMPLEX ON PHOSPHOINOSITIDE DOCKING SITES



Receptor activation can lead to the production of **modified phospholipid molecules** in the adjacent plasma membrane, and these lipids then **recruit specific** intracellular signaling proteins to this region of membrane.

All such signaling complexes form only **transiently** and rapidly disassemble after the extracellular ligand dissociates from the receptor.

Cell-surface receptors relay extracellular signals *via* intracellular signaling pathways



Intracellular signaling molecules can relay, amplify, integrate and distribute the incoming signal.

Participants of the intracellular signaling cascades:

- extracellular signals (soluble ligands, other cell surface receptors)
- Receptors
- Adaptor proteins
- Second messengers (intracellular soluble ligands)
- Effector proteins

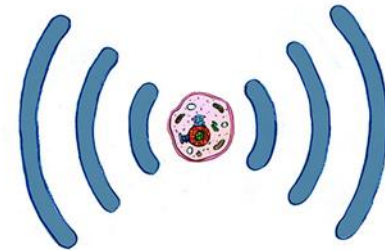
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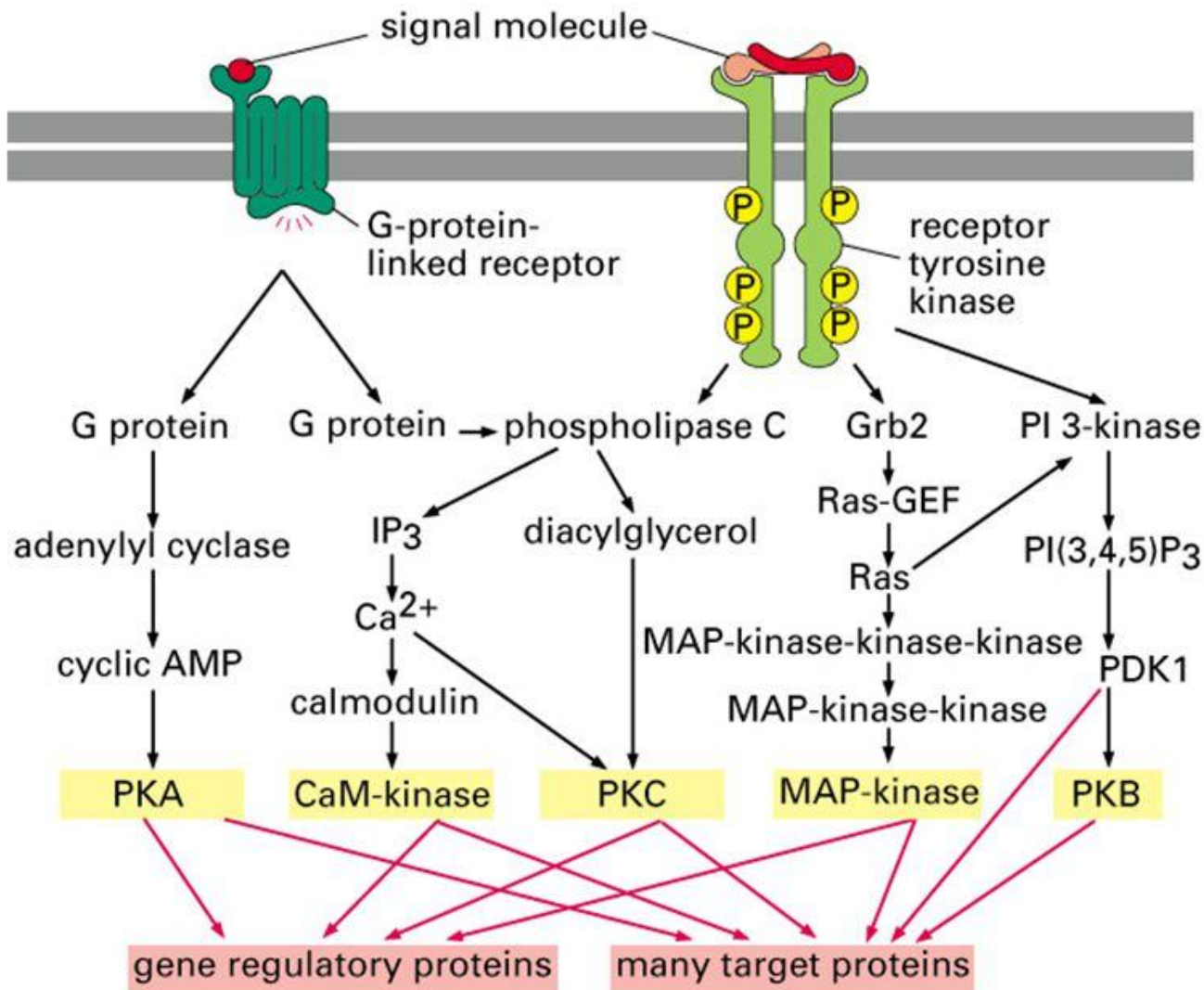
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Cell Signaling a Complex Network

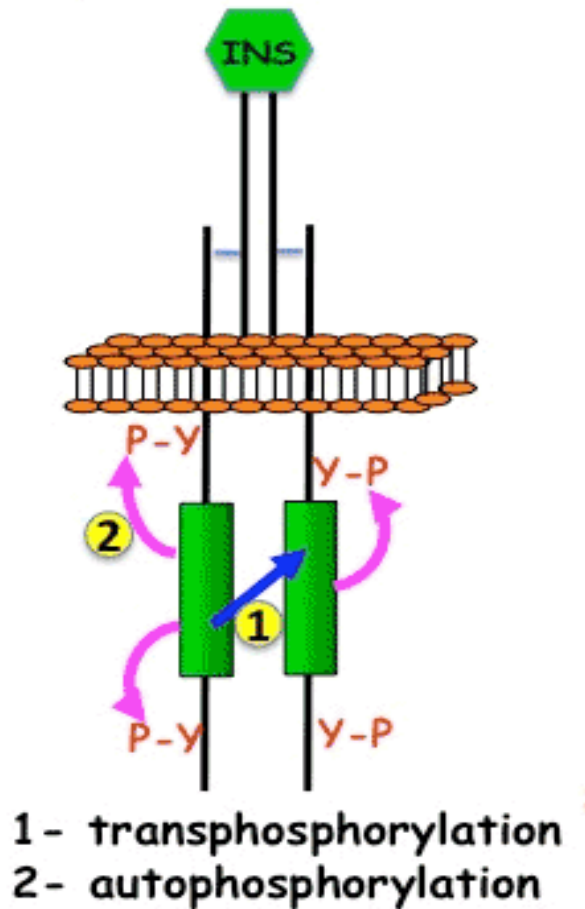


Bifurcation proteins spread the signal from one signaling pathway to another.

Integrator proteins receive signals from two or more signaling pathways and integrate them before relaying a signal onward.

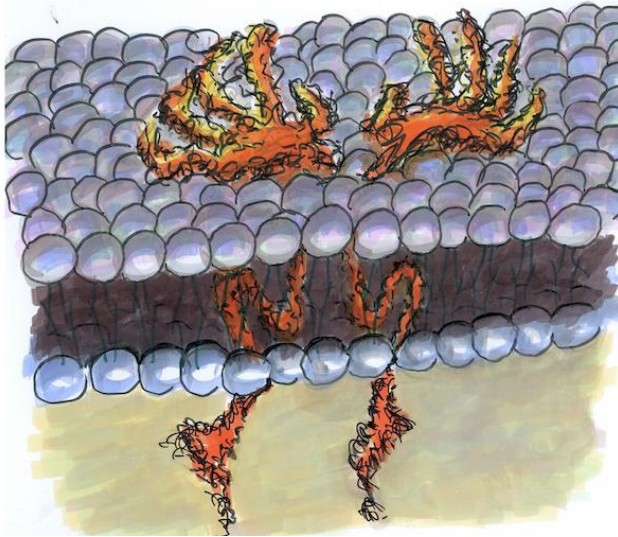
Exemple: insulin receptor (INS-R) signaling pathways

INS-R Activation



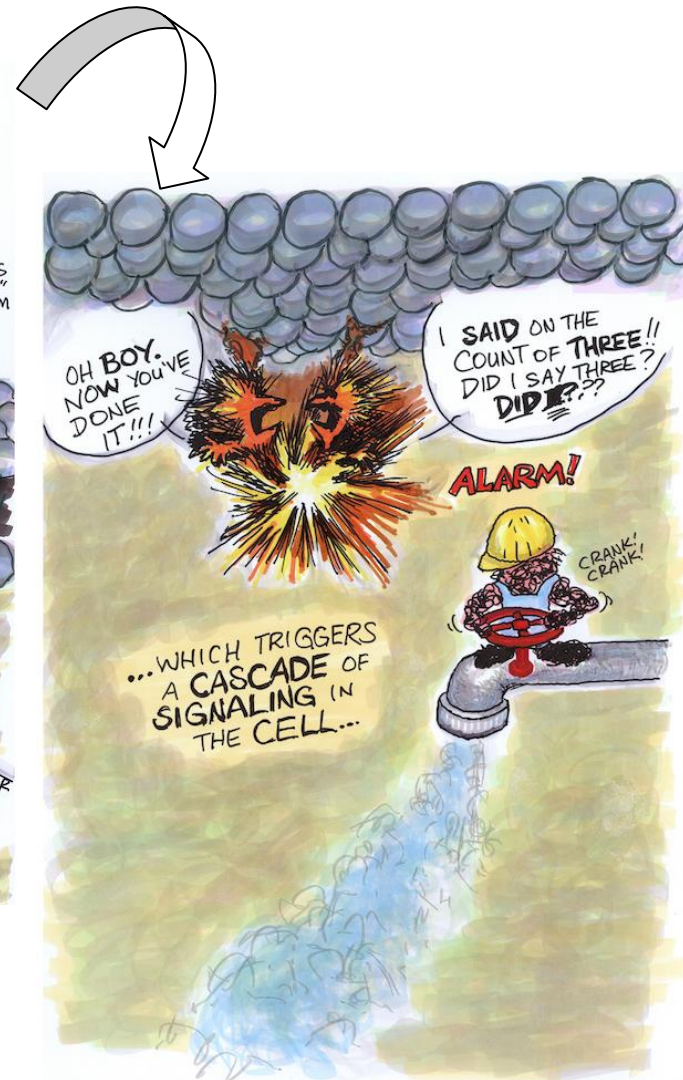
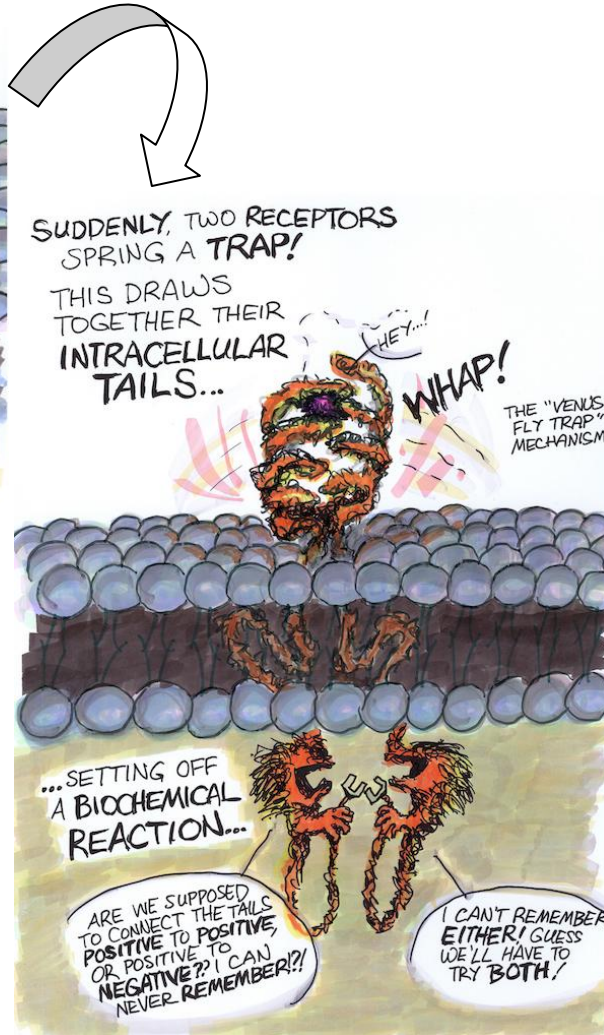
UNSUSPECTINGLY,
A SIGNALING
MOLECULE
APPROACHES THE
SURFACE OF A CELL...

Insulin



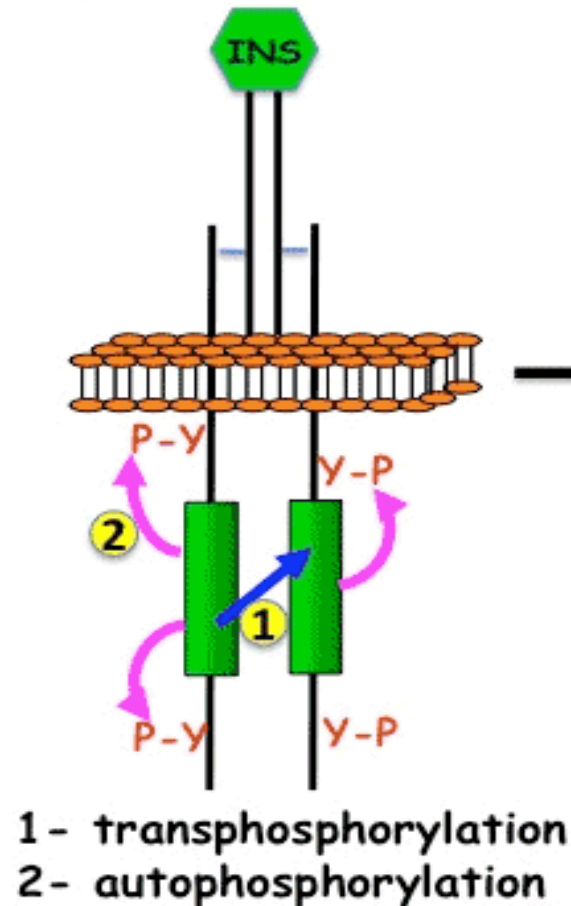
Insulin Receptor

Exemple: insulin receptor (INS-Rc)

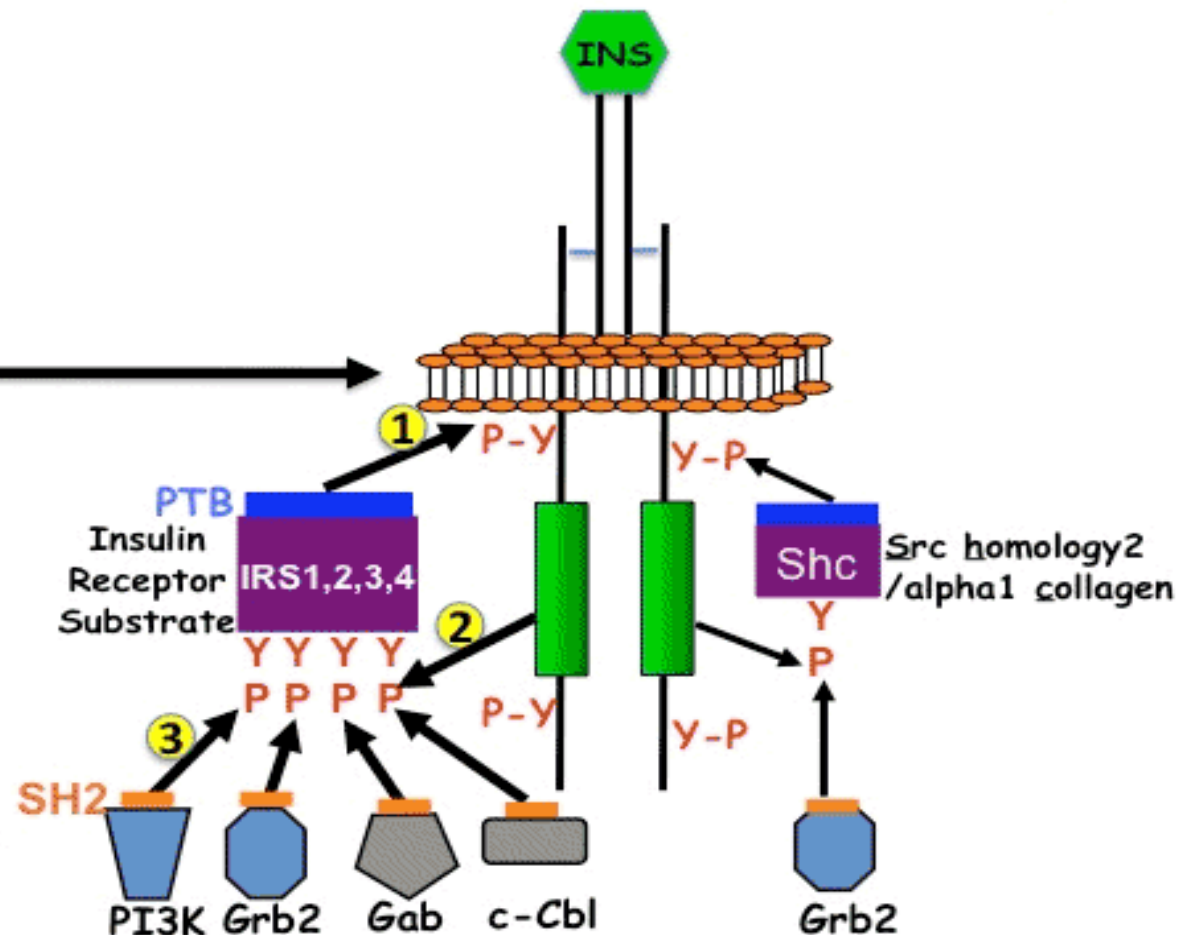


Example: insulin receptor (INS-R) signaling pathways

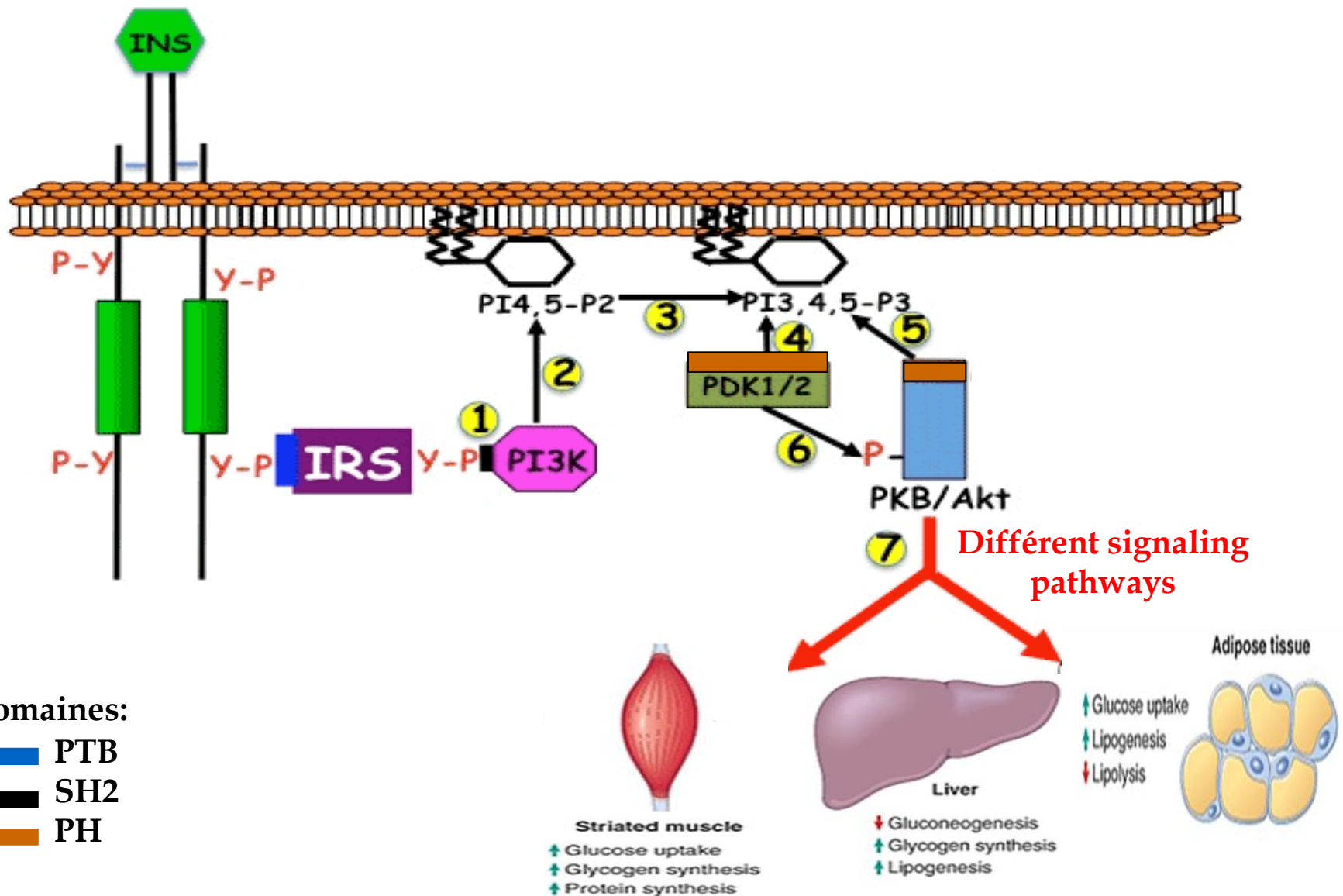
INS-R Activation



Recruitment of signaling partners

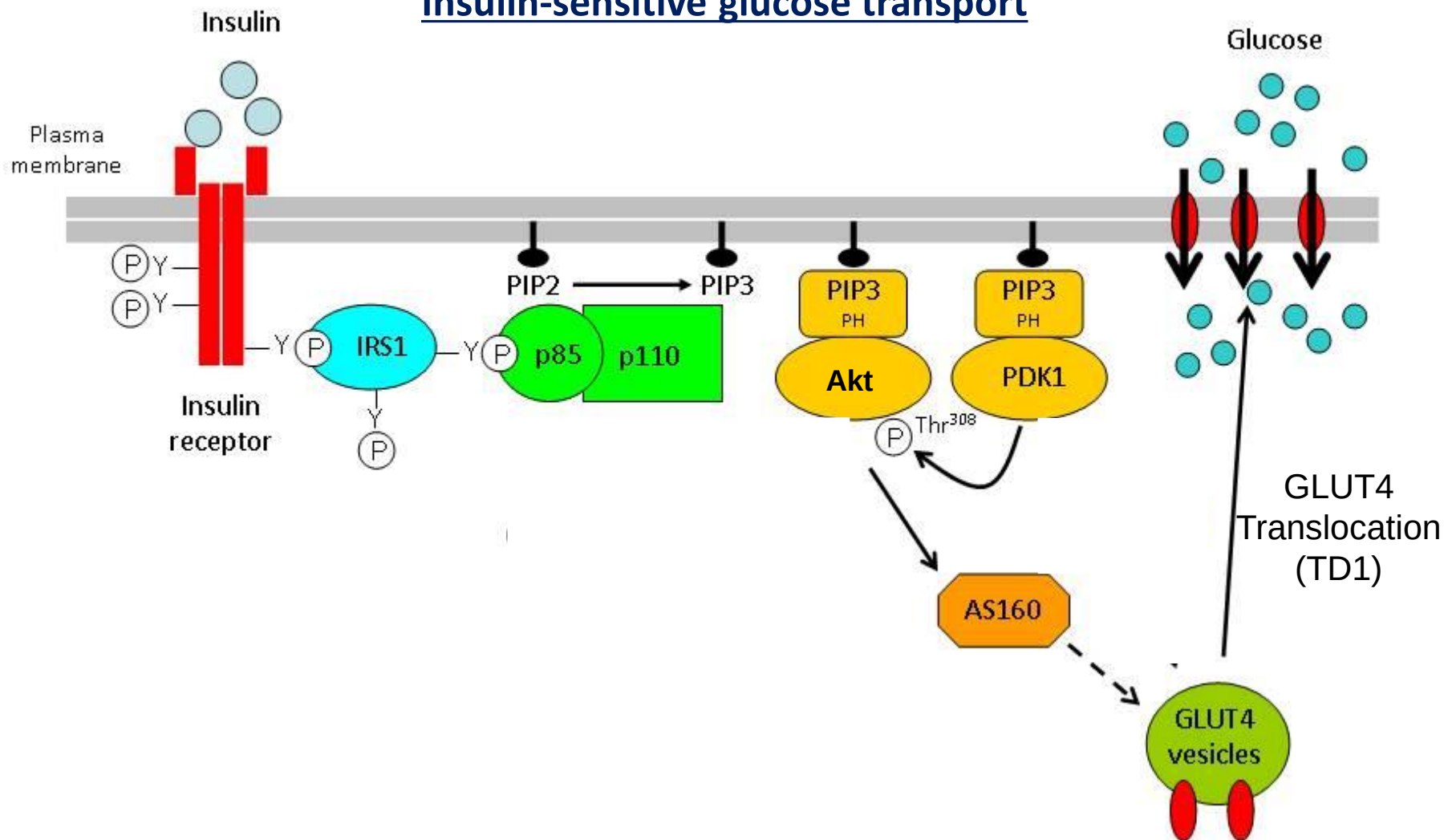


Insulin receptor (INS-R) signaling: metabolic pathway

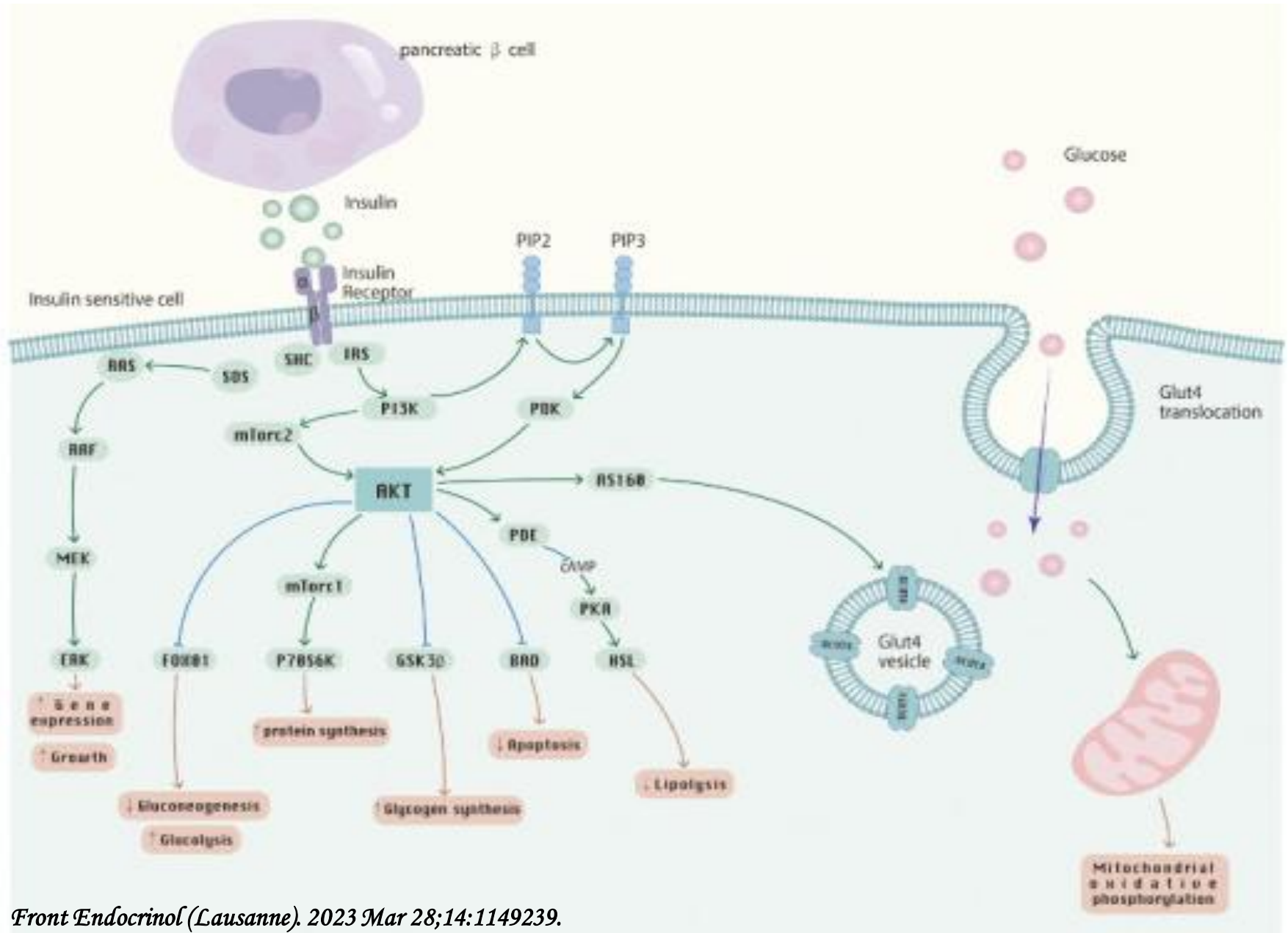


Insulin receptor (INS-Rc) signaling: metabolic pathway

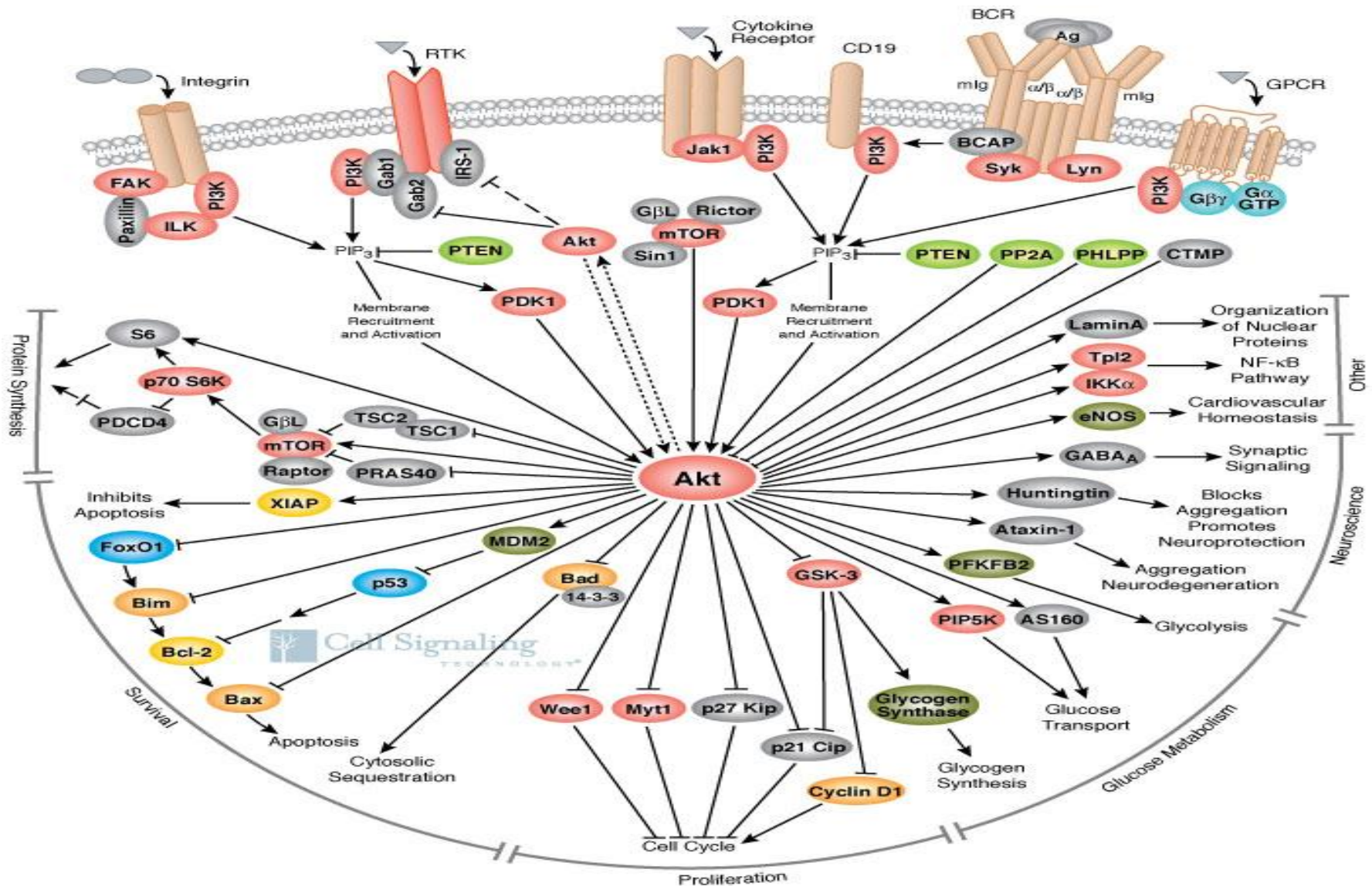
Insulin-sensitive glucose transport



The insulin signaling pathway



Akt: Integrator/Bifurcation signaling protein



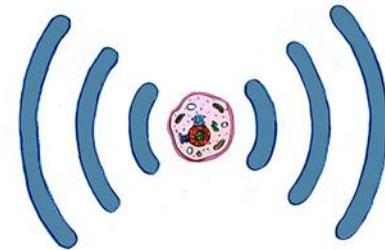
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Insulin resistance

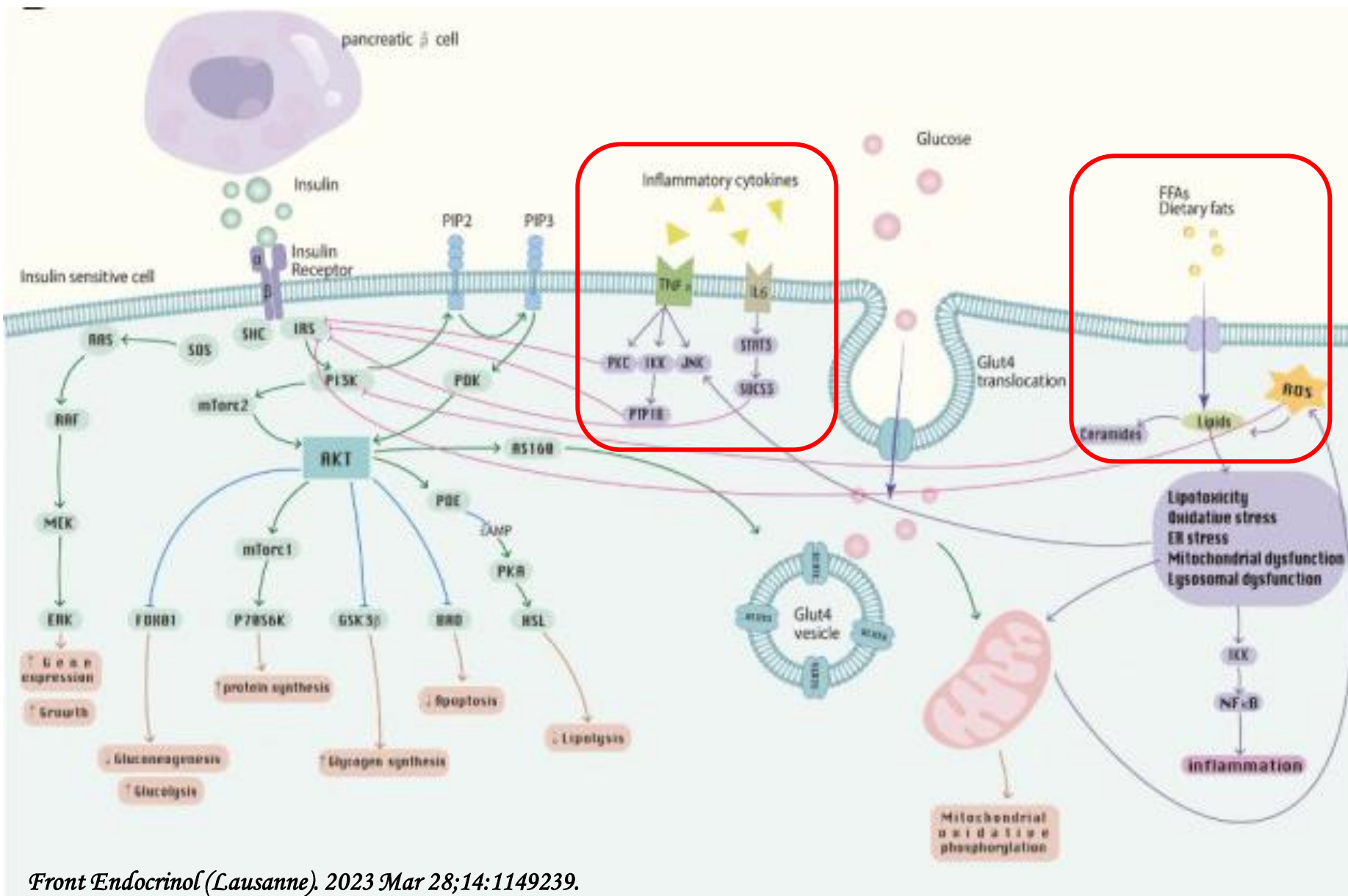
Insulin resistance is one of the most prominent **metabolic disorders**.

It is traditionally defined as **suboptimal action** of a **physiological circulating concentration** of insulin to control **carbohydrate metabolism**.

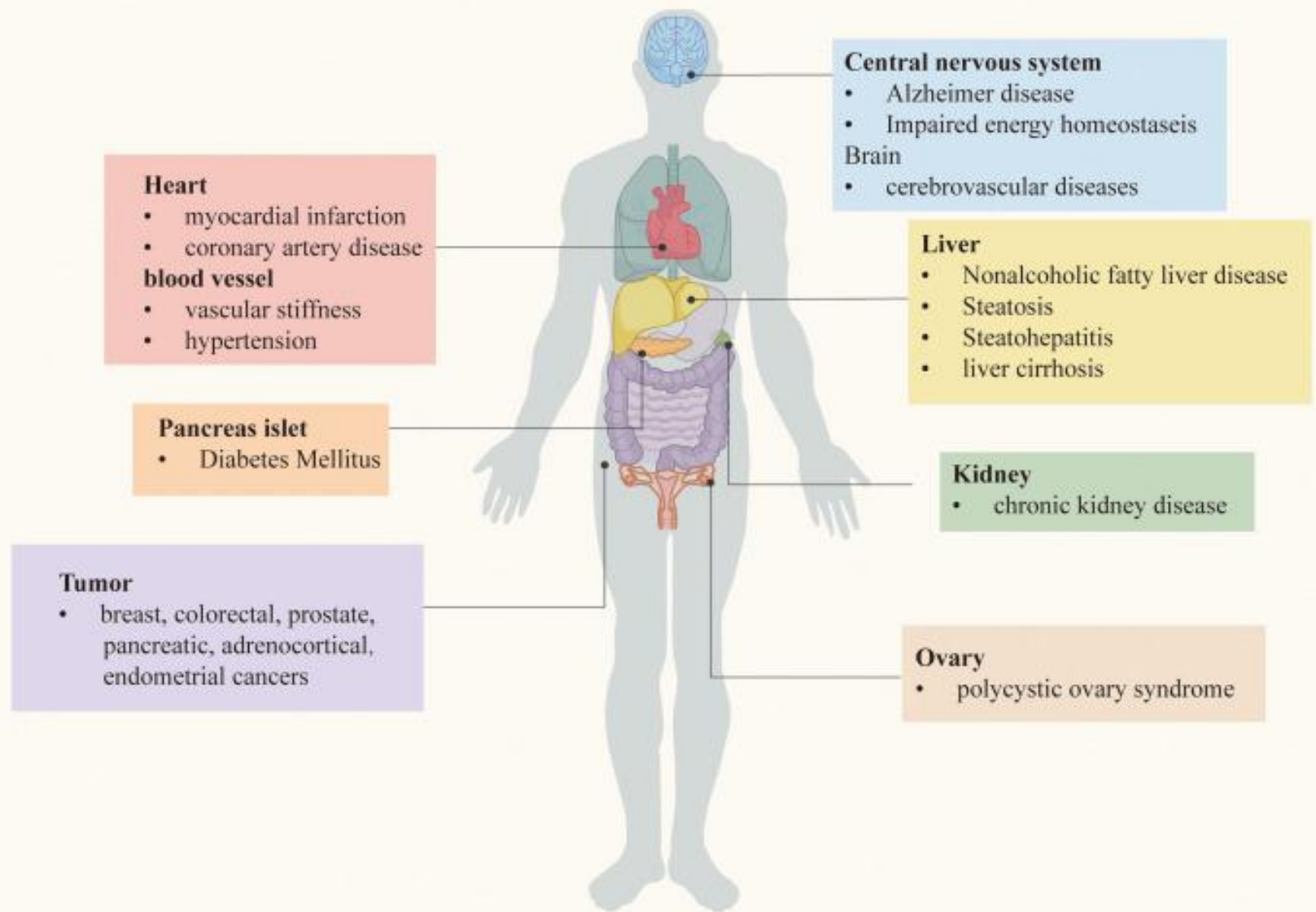
Since insulin has **various functions** in humans, compensatory **hyperinsulinemia** that may result from an inability to control carbohydrate metabolism usually has **adverse physiological outcomes** in other organs and systems.

Insulin resistance thus constitutes a common underlying pathophysiological condition for many **clinical disorders**.

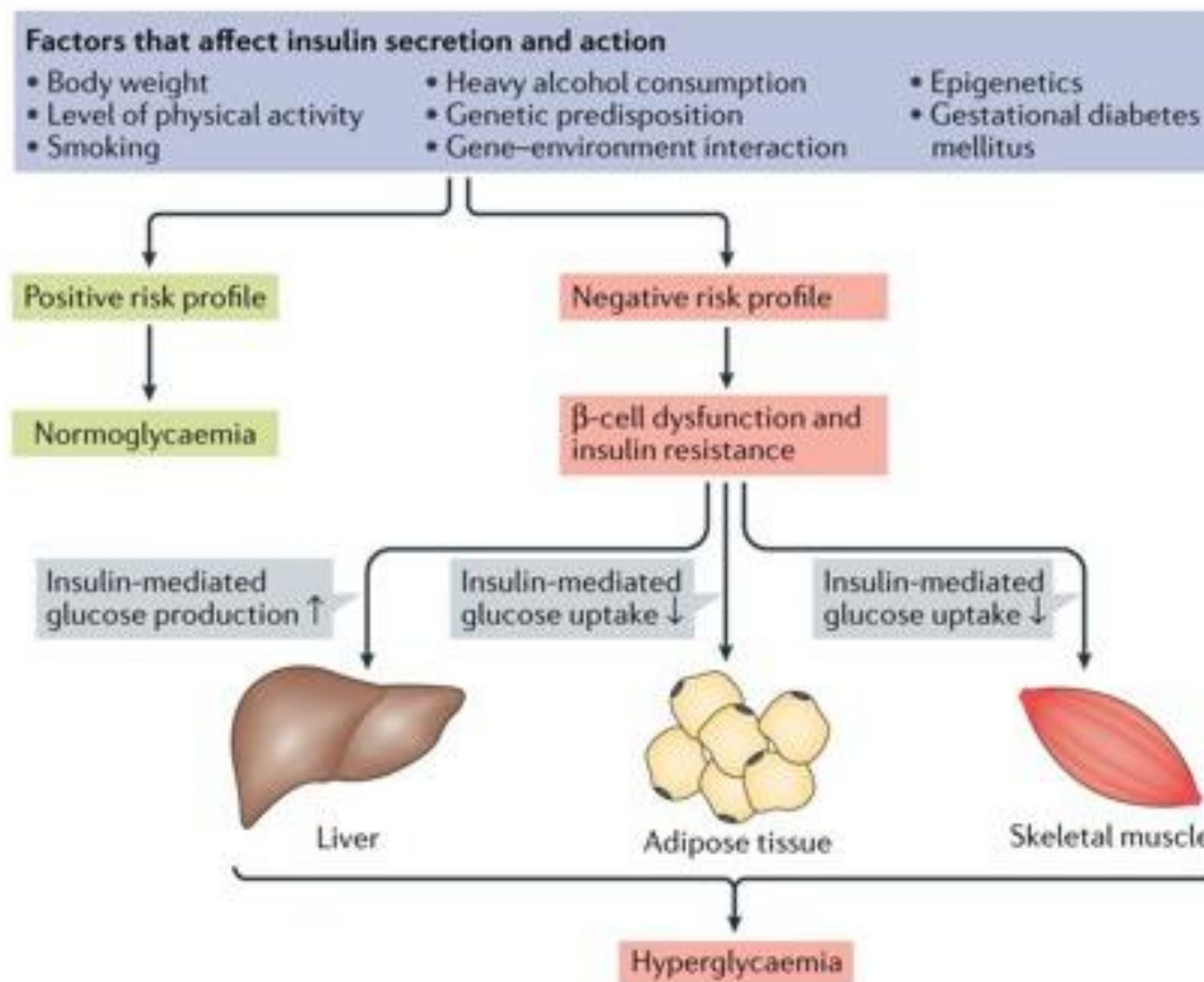
Abnormalities in the insulin signaling pathway caused by inflammatory cytokines, FFA, etc., involved in the development of IR.



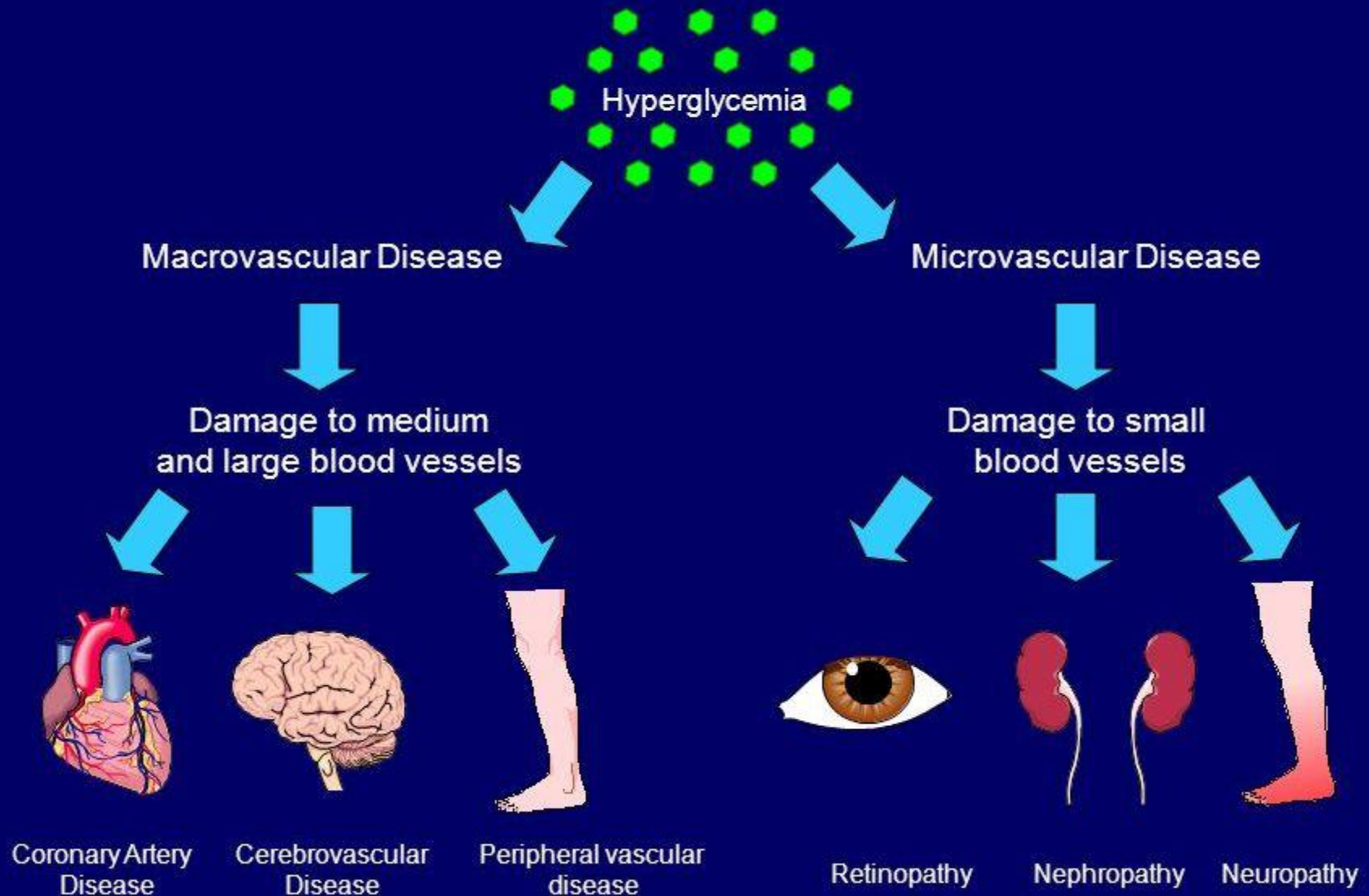
Chronic metabolic diseases may be induced by IR

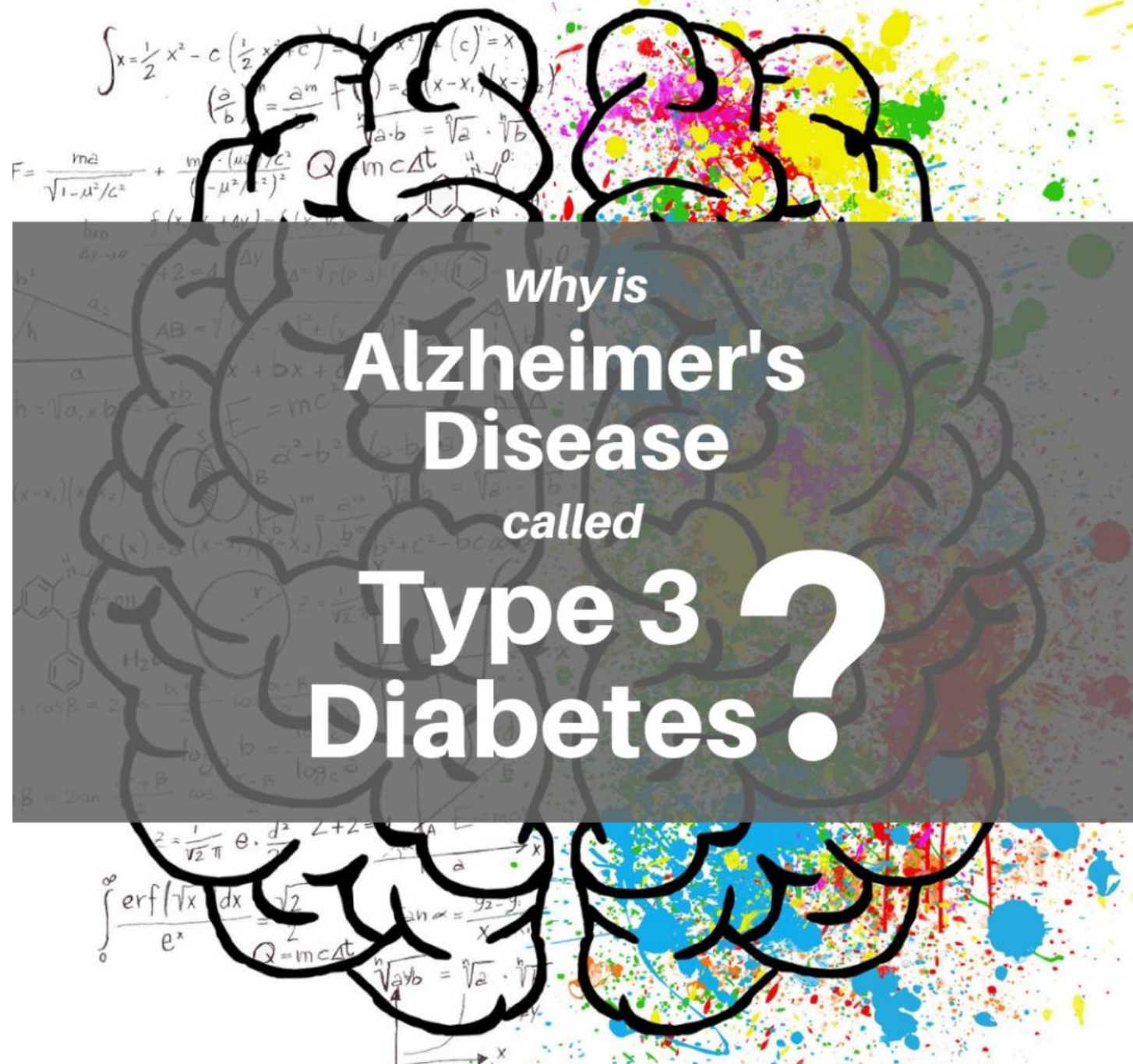


Pathophysiology of hyperglycemia in type 2 diabetes mellitus

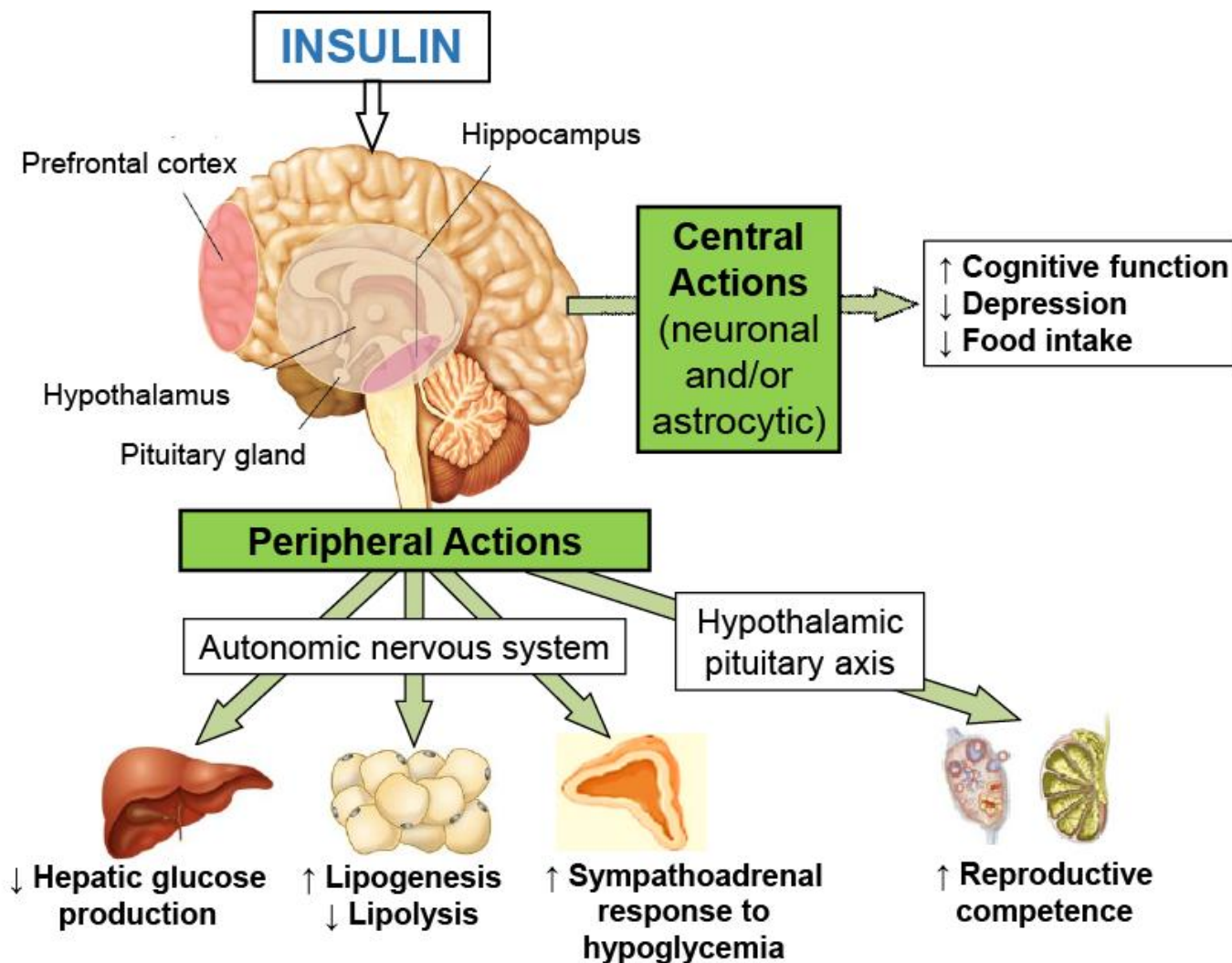


Long-term Complications of Type 2 Diabetes





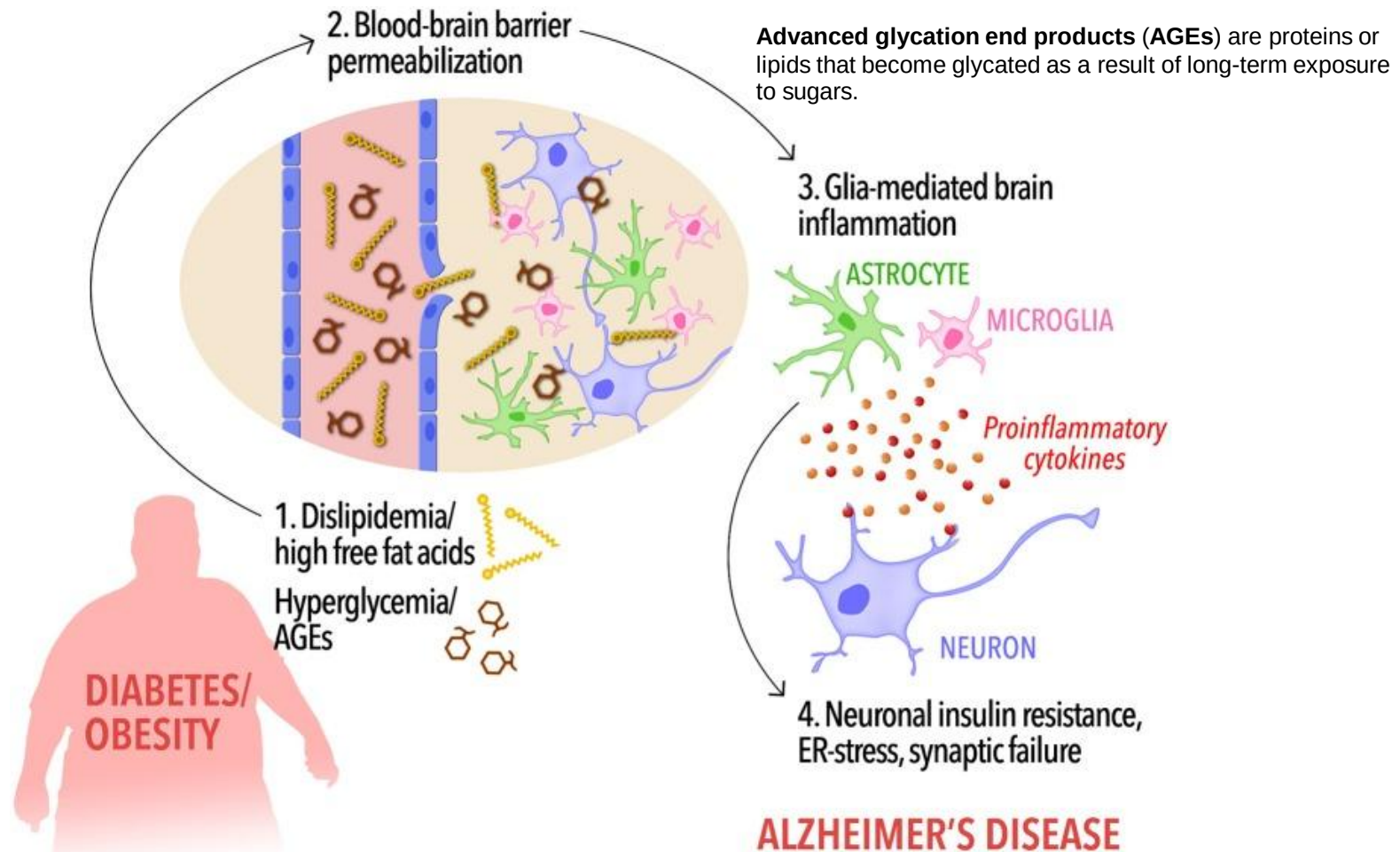
Why is
**Alzheimer's
Disease**
called
**Type 3
Diabetes ?**



Insulin Resistance in Alzheimer's Disease

- type 2 diabetes doubles the risk for developing Alzheimer's disease. This is likely due to the fact that **in both type 2 diabetes and Alzheimer's disease, there is a deficiency of insulin signaling**.
- Insulin resistance may be a **leading cause of dementia**, as this glucose metabolism deficiency in the brain contributes to symptoms such as **loss of memory, decrease in judgment and reasoning skills**.
- Loss of brain cell energy means that the brain can no longer carry out **memory** and **cognitive functions** normally and also can not produce the parts of brain cells needed to replace those that wear out over time leading to **degeneration of the brain itself**.

Insulin Resistance in Alzheimer's Disease

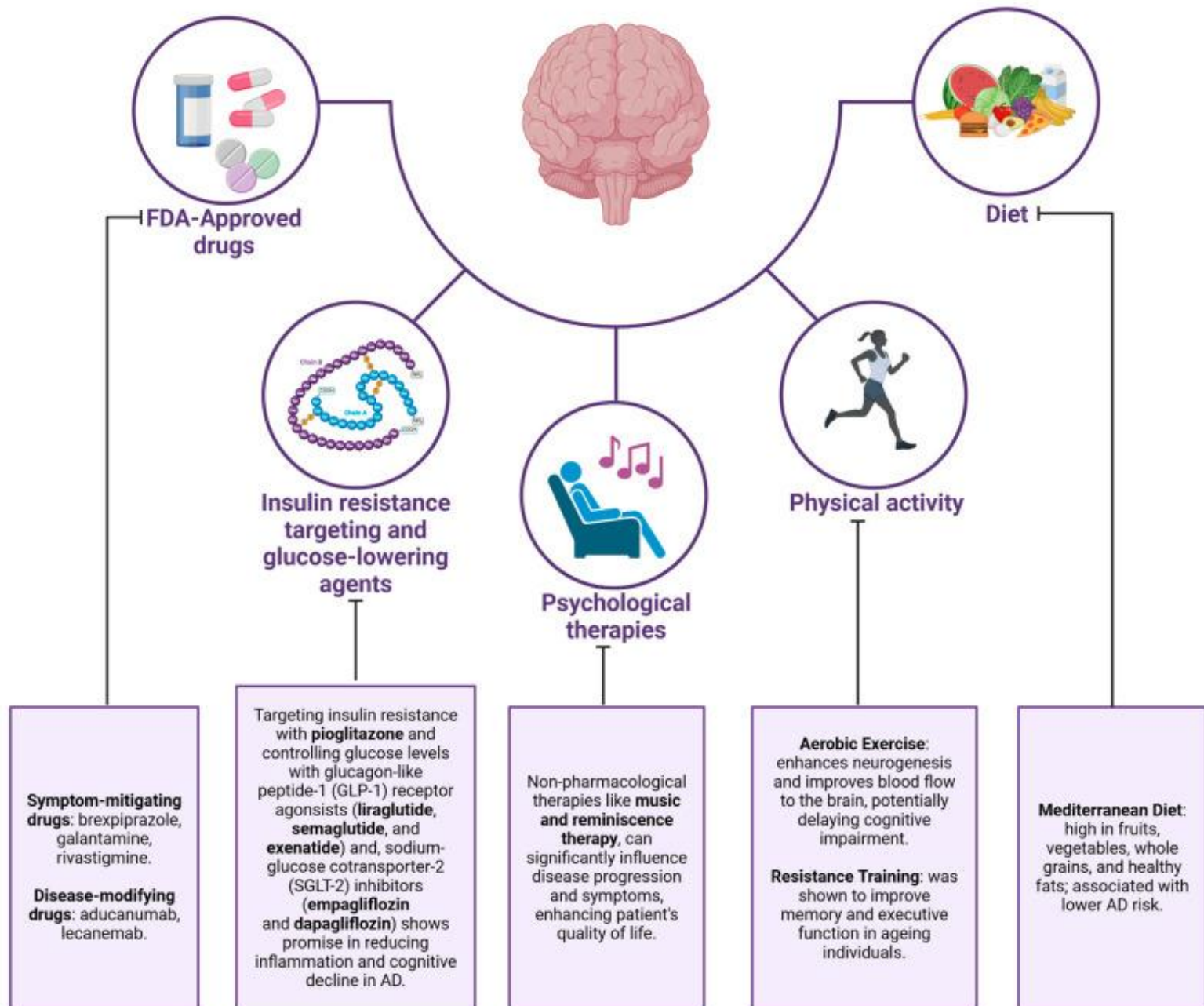


[Front Neurosci.](#) 2018; 12: 830

A possible cascade of events connecting peripheral metabolic dysregulation to dementia. In diabetic and/or obese subjects, dyslipidemia, and increased circulating free fat acids as well as hyperglycemia and elevated peripheral AGEs levels (1) may increase blood-brain barrier permeability, allowing the influx of FFAs into the brain (2). Disrupted BBB along with high levels of brain FFAs and AGEs, in turn, would cause activation of microglia and astrocytes and the release of proinflammatory cytokines (3). Low-grade, chronic brain inflammation leads to detrimental events in neurons, including insulin resistance (4), priming the brain to cognitive impairment and Alzheimer's disease.

Current treatment options for Alzheimer's disease (AD)

Int J Mol Sci. 2024 Nov 7;25(22):11955.





The Mediterranean diet concept and definition

1. Keys A, Menotti A, Karvonen MJ, Aravanis C, Blackburn H, Buzina R, et al. The diet and 15-year death rate in the seven countries study. Am J Epidemiol. 1986; 124: 903–15.



- It is not a “diet” in the restrictive sense, but a well-balanced eating pattern you can customize and use to make healthier choices that honor your body.
- The MedDiet, the term that was coined by Ancel Keys back in 1960 [1], is one of the most studied and well-known dietary patterns worldwide.
- The origins of the traditional MedDiet pattern are found in civilizations surrounding the Mediterranean Sea, so that this pattern has been closely associated with the social behaviors and lifestyles of that region.
- The MedDiet has been recognized by UNESCO as an intangible cultural heritage that is deeply rooted into its geographical origin and whose agricultural and dietary practices have a responsible interaction with the environment.

The Mediterranean diet and health: a comprehensive overview

Mediterranean diet pyramid: a lifestyle for today
guidelines for adult population

Serving size based on frugality
and local habits



Wine in moderation
and respecting social beliefs

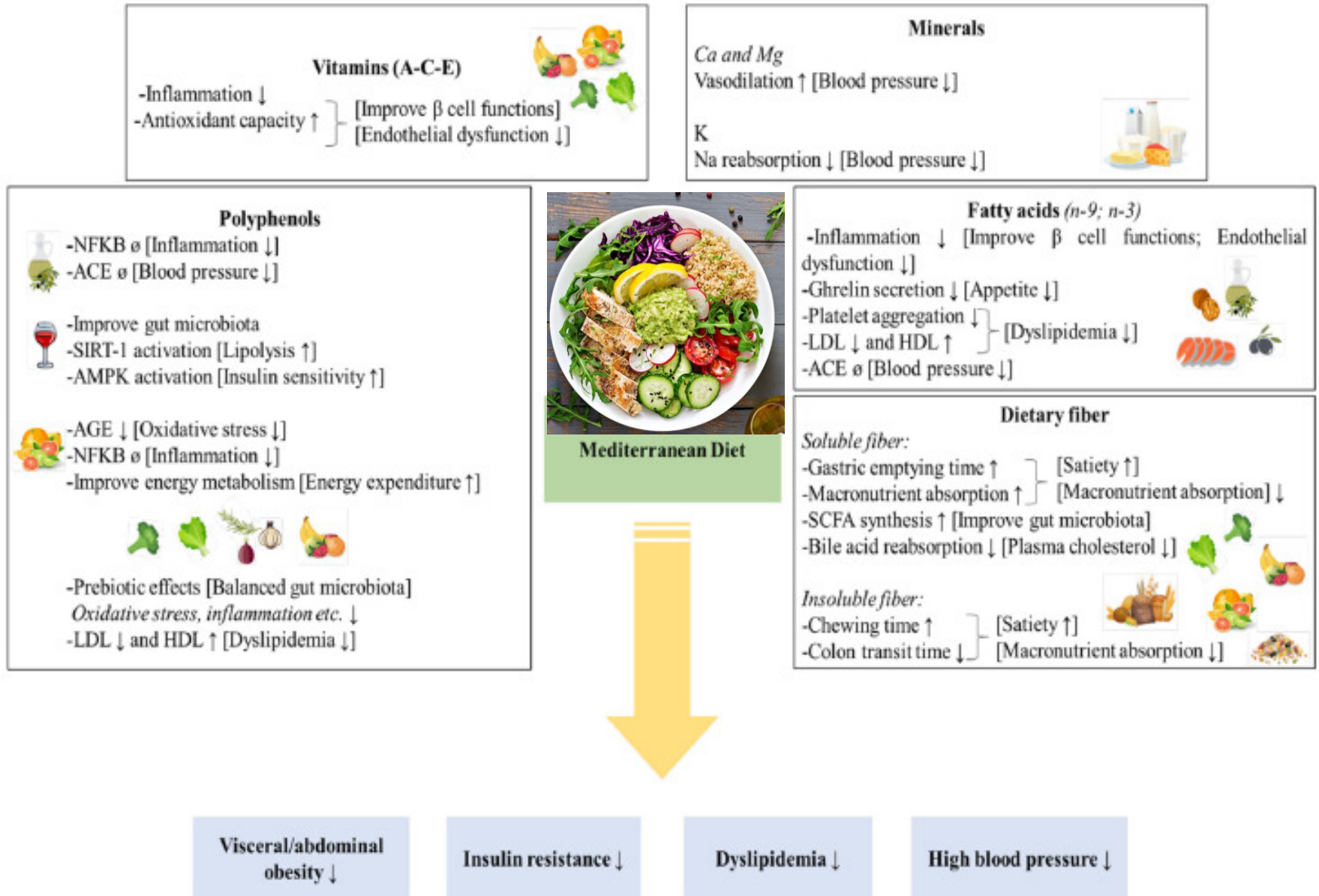


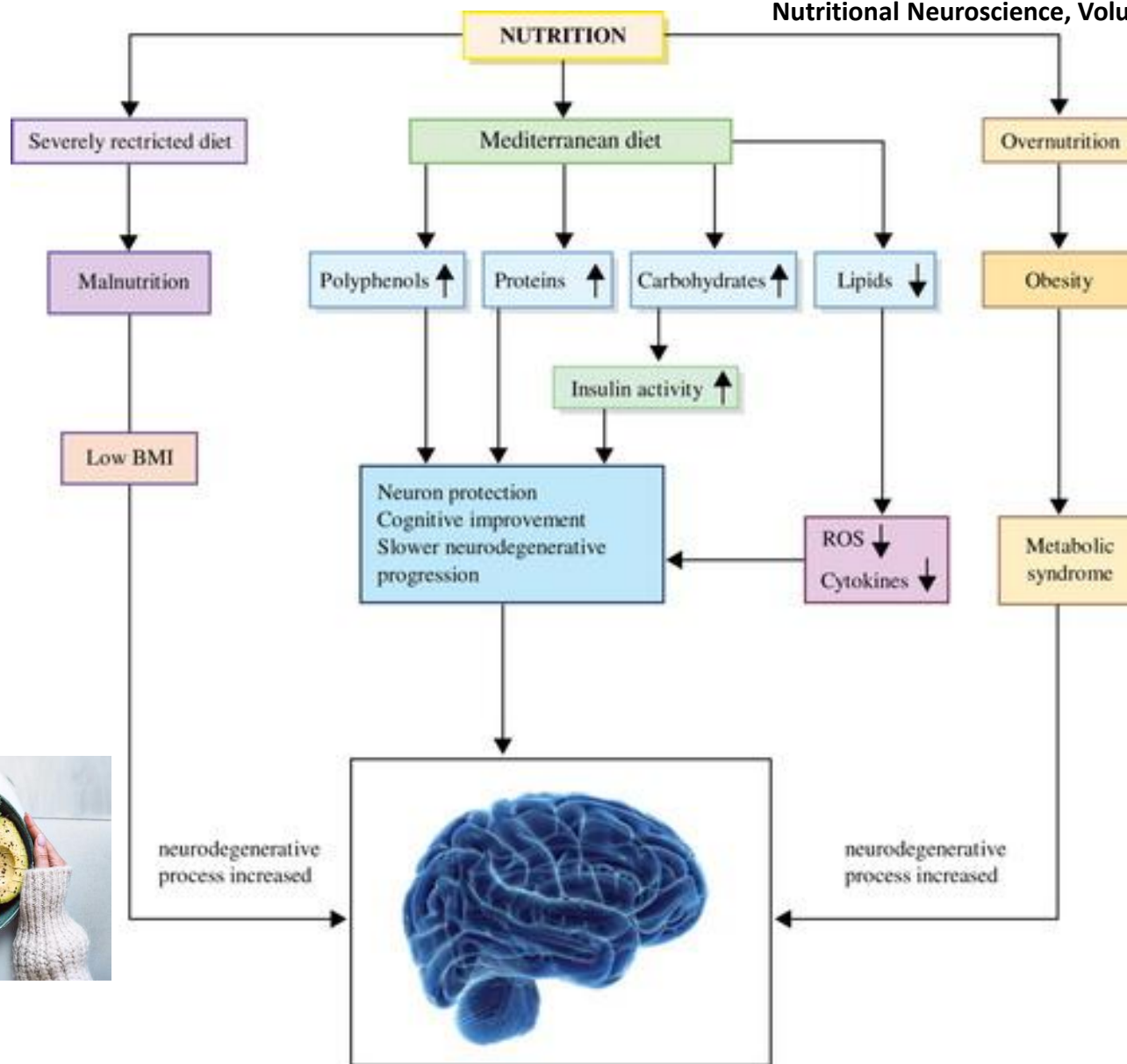
© 2010 Fundacion dieta mediterranea the use and promotion of this pyramid is recommended without any restriction



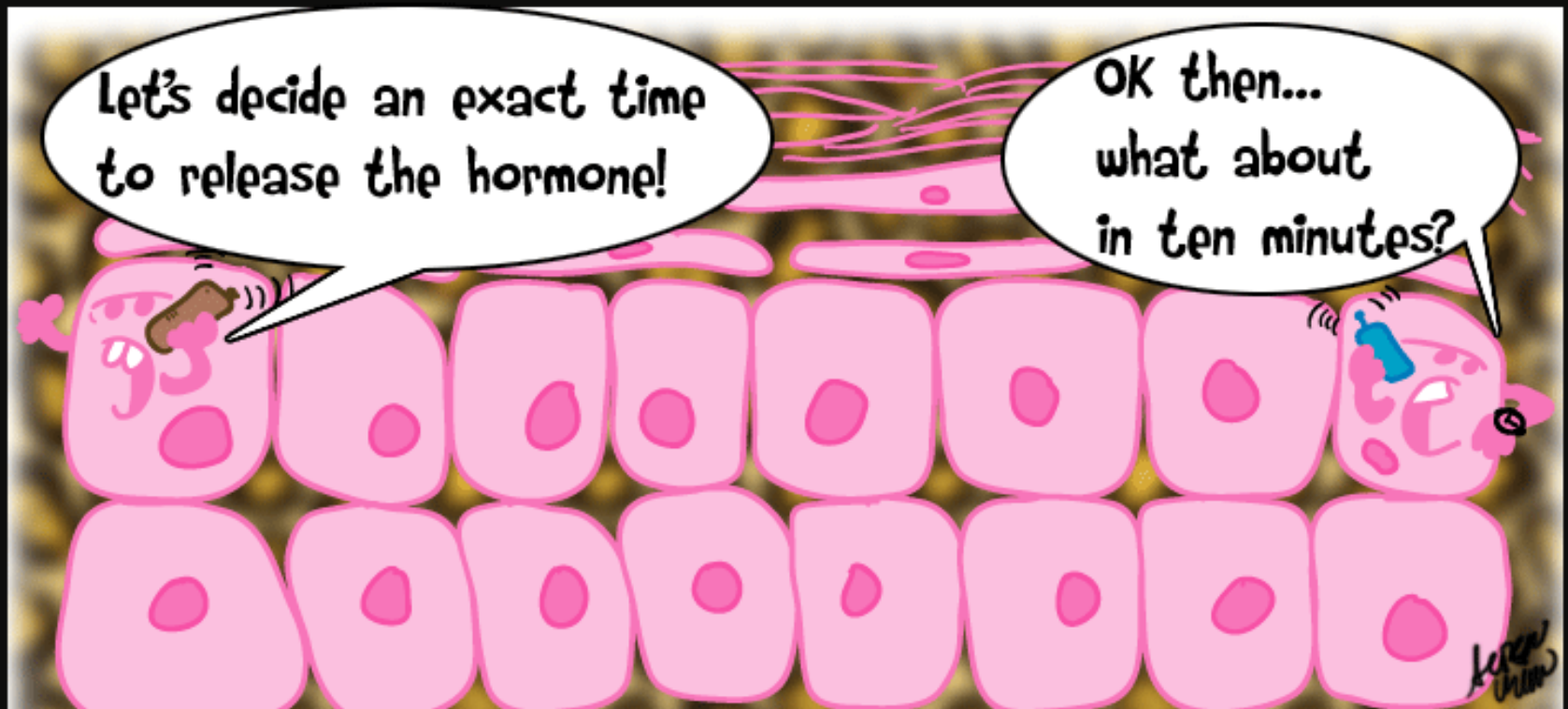
Fundación
Dieta Mediterránea

Effects of the MedDiet on different components of metabolic syndrome.





Mediterranean diet, due to the controlled caloric intake, polyphenol and flavonoids (fruit, vegetables, olive oil, red wine) plus a regular intake of protein (fish and meats) and carbohydrates (cereals) reduces the insulin resistance and the pro-inflammatory cytokines level determining neuronal protection and slowing neurodegenerative progression.



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